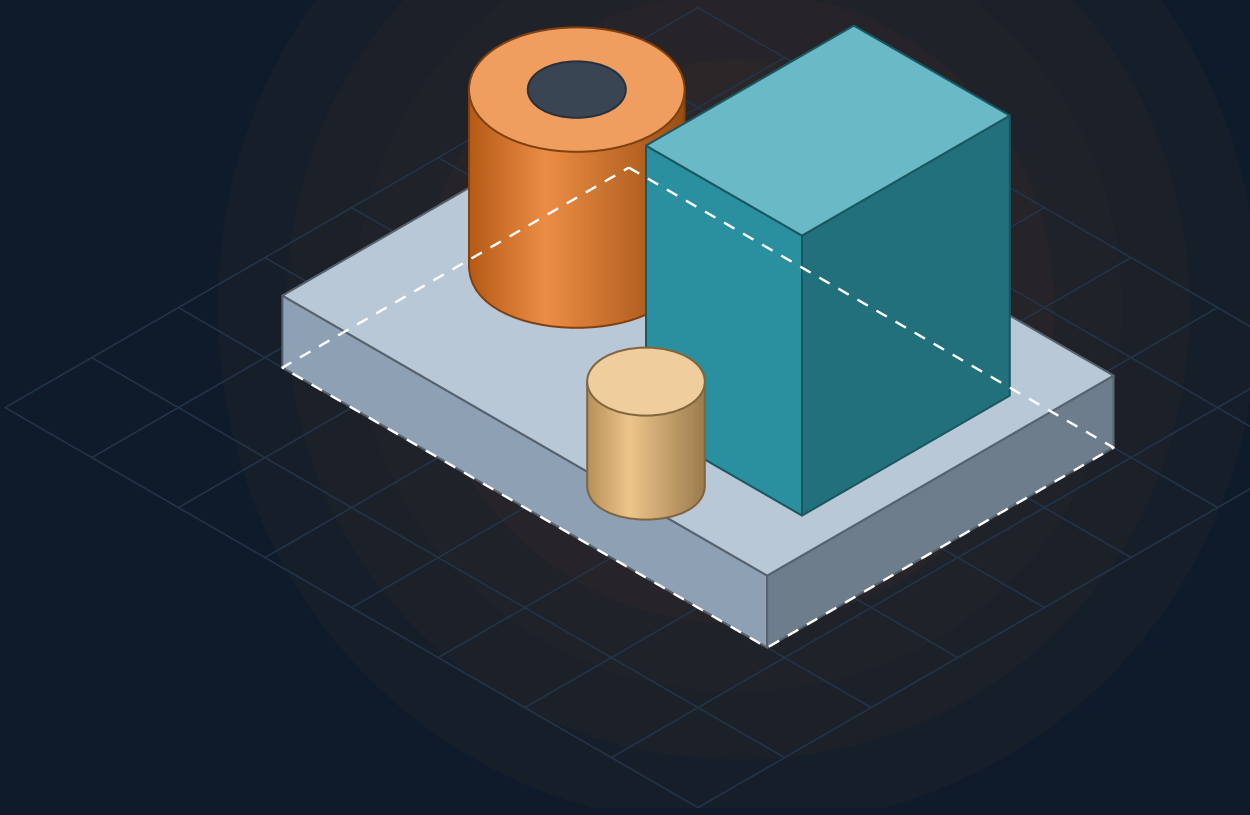
THE HOBBYIST'S ILLUSTRATED GUIDE

Autodesk Fusion

Every modeling feature, menu and tool, explained for makers, tinkerers and weekend designers.



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1

PART 1

Meet Fusion

Fusion is a single program that takes an idea from a rough sketch all the way to a 3D-printed or machined part. This part explains what it is, what the free hobbyist license covers, and how to find your way around the screen.

Chapter 1 · Why hobbyists love Fusion Chapter 2 · The free personal-use license Chapter 3 · Getting started Chapter 4 · A tour of the interface Chapter 5 · Navigating the 3D canvas

CHAPTER 1

Why hobbyists love Fusion

Autodesk Fusion (still called "Fusion 360" by many people) is a cloud-connected 3D CAD program, which means it designs, tests and prepares real objects for manufacture. This guide walks through every modeling feature a hobbyist is likely to meet and explains what each menu and tool does, in plain language.

What Fusion actually is

CAD stands for **computer-aided design**. Where a drawing program makes pictures, a CAD program makes *objects with real dimensions*: a hole is exactly 5 mm wide, a wall is exactly 2.4 mm thick, and two parts either fit together or they do not. Fusion combines several kinds of software that used to be sold separately:

- **CAD**: sketching and 3D modeling of solid, surface, mesh and freeform shapes.
- **CAM**: computer-aided manufacturing, meaning toolpaths for CNC machines and settings for 3D printers.
- **CAE**: engineering analysis such as stress testing, so you can see whether a bracket will break before you print it.
- **Presentation tools**: photo-realistic rendering, exploded-view animations and 2D technical drawings.
- **Electronics**: schematics and circuit-board layout for projects that need a PCB.

Autodesk's own help pages describe this as capabilities grouped into purpose-driven **workspaces**: Design, Electronics, Generative Design, Render, Animation, Simulation, Manufacture and Drawing. You switch between them from a drop-down at the top-left of the toolbar, and your model comes along with you.

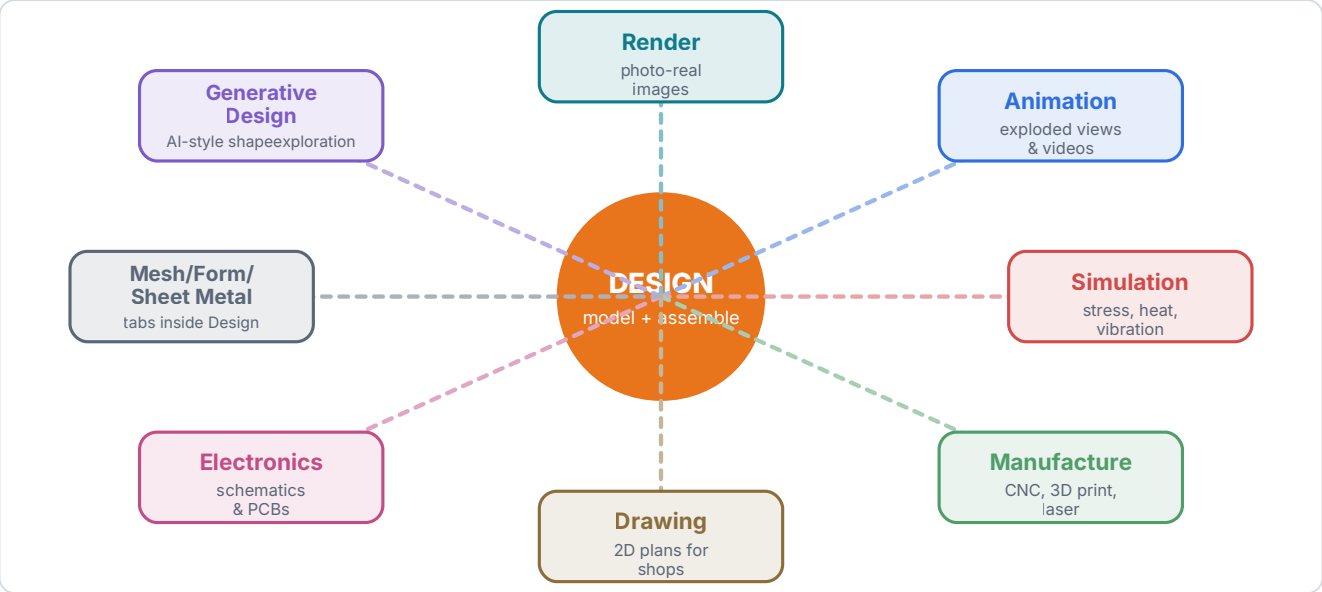


Figure 1. The eight workspaces in Fusion, arranged around Design, where most hobby work happens. The Mesh, Form (T-Spline), Sheet Metal and Plastic tools live inside the Design workspace rather than in workspaces of their own.

The ten reasons it suits hobbyists

Strength	What it means for you
1. Free for personal use	Qualifying hobbyists get a free license for home-based, non-commercial projects (details in Chapter 2).
2. History-based (parametric) modeling	Fusion remembers each step in a **timeline**. Made a hole in the wrong place? Click it in the timeline, change the number, and the model rebuilds itself.
3. All-in-one	You can sketch, model, render, check stress, and slice for a printer without exporting to five different programs.
4. Works on modest computers	Because it is cloud-connected, heavier jobs such as some simulation and rendering can run on remote servers on plans that include them. Core modeling runs on your own machine.
5. Cloud storage and version history	Every save is a new version. Nothing is lost to a crash, and you can restore yesterday's design with two clicks.
6. Many ways to model	Parametric solids, direct edits, surfaces, T-Spline sculpting and mesh editing live side by side, so you can pick the easiest tool for each job.
7. Friendly dialogs with live previews	Nearly every command shows a preview in the 3D view while you type values, plus little manipulator arrows you can drag.
8. Direct path to the real world	One click sends a design to a 3D-print slicer; the Manufacture workspace creates toolpaths for CNC routers and mills; sheet-metal designs unfold to a flat pattern for laser cutting.
9. Huge learning ecosystem	Official tutorials, a very active user forum, countless video courses and a built-in assistant (Chapter 42) mean you are never stuck for long.

Strength	What it means for you
10. It grows with you	The same skills carry from a first coaster to complex assemblies, so time spent learning keeps paying off.

KEY IDEA

The core idea: in Fusion you almost never "draw" a 3D shape directly. You **sketch** a flat outline, turn it into a **feature** (extrude it, revolve it, sweep it), then **refine** it (fillet, hole, shell). Each step is stored in the timeline so it can be edited later.

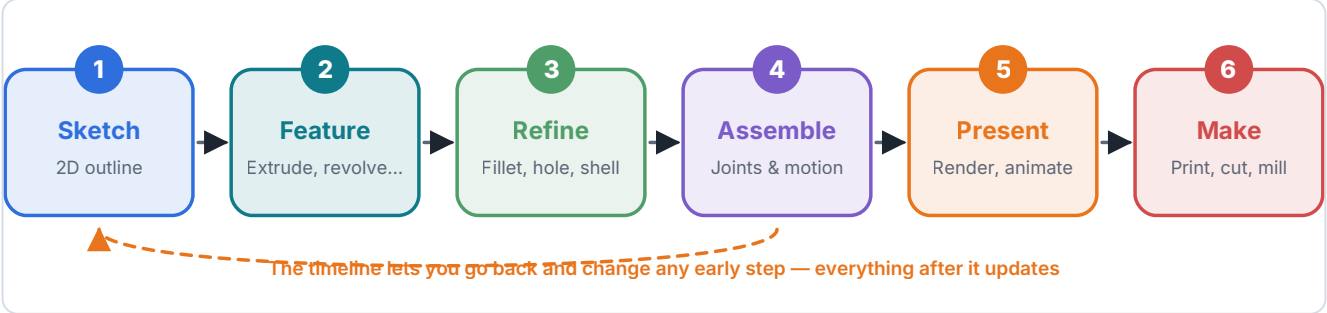


Figure 2. The usual Fusion workflow. Nearly every hobby project, from a coaster to a drone frame, follows these six steps.

What this guide covers

The guide is organized to match how you will learn. **Part 1** orients you. **Part 2** explains the ideas under the hood (timeline, components, parameters). **Part 3** is sketching. **Part 4** is the heart: every Create, Modify, Construct, Inspect and Insert tool in solid modeling. **Part 5** covers the other modeling styles: surface, T-Spline, mesh, sheet metal and plastic. **Part 6** covers assemblies and motion. **Part 7** covers rendering, animation and drawings. **Part 8** covers simulation, generative design, CAM, 3D printing and electronics. **Part 9** turns it all into hands-on projects, shortcut cheat sheets, a learning roadmap and a glossary.

TIP

You do not need to read straight through. If you are brand new, read Parts 1 to 3 and then jump to the project walkthroughs in Part 9. Return to Part 4 as a reference whenever you meet a tool you do not recognize.

CHAPTER 2

The free personal-use license

Fusion is a commercial product, but Autodesk offers a free version for hobbyists. Knowing exactly what it includes (and what it does not) saves frustration later.

Who qualifies

According to Autodesk's own description, **Fusion for personal use** is a free version designed for home-based, non-commercial projects. It is intended for hobbyists, makers and DIY enthusiasts, and for people who generate less than **$1,000 USD per year** from their work. Selling products, freelancing, consulting or any revenue-earning activity beyond that threshold requires a commercial subscription. Autodesk's licensing pages also describe personal-use access as a multi-year subscription that you renew, and the product pages note that you sign in or create an Autodesk account to download it.

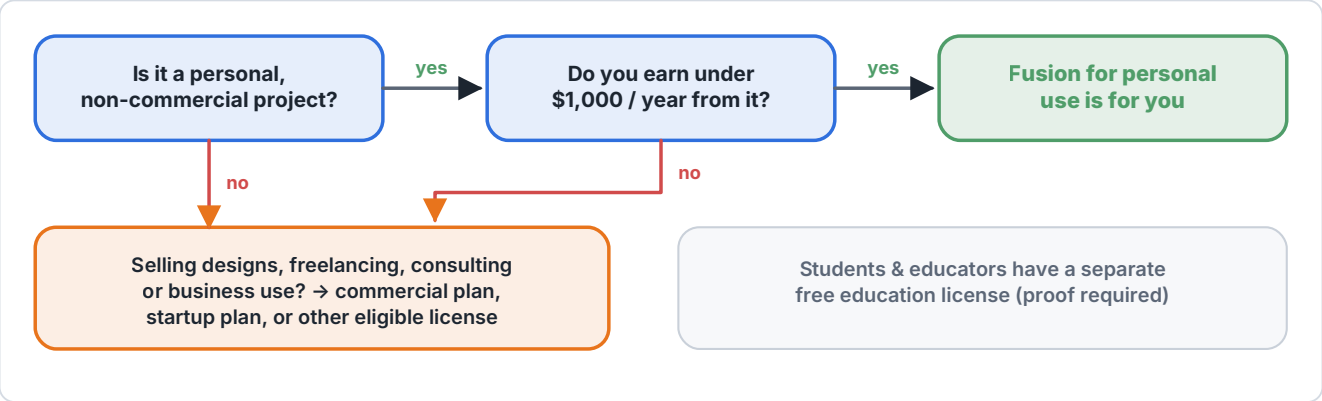


Figure 3. A quick eligibility check. When in doubt, read Autodesk's current terms, because licensing details can change.

What you get

Autodesk's comparison lists the following as included in the personal-use version:

- **3D modeling**: parametric, direct, surface, mesh and freeform design tools. This is the core of this guide, and it is all available.
- **Cloud data management**: projects stored in the cloud with version history, accessible from multiple devices.
- **Basic CAM**: toolpath generation with limited functionality.
- **Limited electronics**: PCB and schematic design with restrictions.
- **Learning resources**: tutorials, forums, videos and community content.

What is limited or missing

Area	Personal use (per Autodesk's comparison)
CAM	Limited compared with the full version's comprehensive machining toolkit.
Simulation (CAE)	Limited compared with the full version. Cloud simulation is not included.
Electronics / PCB	Limited, compared with the unified electronics and PCB workflows in the full version.
Collaboration	Single-user only. No multi-user team workflows.
2D drawings	Limited, compared with automated drawings in the paid version.
File types	Limited import and export formats. (Autodesk reversed one proposed restriction in the past: STEP export was kept available after community feedback, as reported by the 3D-printing site Fabbaloo.)
Support	Community forum only.
Configurations	Not included. (Chapter 33 shows a hobbyist alternative using parameters.)
Bill of materials	Not included.
Cloud rendering	Not included. Local rendering still works.
Extensions	Fusion extensions are not available to personal users.

WATCH OUT

Several online forums mention a cap of **10 active, editable documents** on personal licenses, with older projects archived rather than deleted. Autodesk's comparison page does not state this, so treat it as community-reported and confirm in your own account. Generative design availability is also not addressed on that page; check inside the program or on Autodesk's site before planning a project around it.

Other free routes

Eligible students and educators can get a free **education license** (proof required) with fewer restrictions, and Autodesk offers a **30-day trial** with full access, which is not the same thing as the personal-use license. Autodesk's page lists the commercial price as $680 per year or $85 per month at the time of research (October 2026); check its pricing page for current figures.

HOBBYIST ANGLE

The limitations hit hobbyists less than you might expect. Hobby designs are mostly parts and assemblies, and the entire modeling toolkit is included. If you outgrow the free tier, you will know exactly which feature you are missing.

CHAPTER 3

Getting started

Installing Fusion, signing in, understanding where your files live and saving a first design.

Install and sign in

1. Create or sign in to a free Autodesk account. Use the same email you will use for Fusion every time.
2. From Autodesk's personal-use page, choose the personal-use download. The installer also activates the license for new personal-use accounts.
3. Run the installer and launch Fusion. The first start-up asks you to sign in and may take a minute to finish loading.
4. You land on **Home**, a start page that lists your recent designs, projects, and learning links. (Autodesk's April 2026 update note says new users start in Home by default.)

Fusion runs on Windows and macOS, and there is a web client for viewing and light editing. A reasonably recent computer with a dedicated or modern integrated graphics chip is plenty for hobby work.

Where your files live: hubs, projects and folders

Fusion stores designs in the cloud instead of in loose files on your desktop. This takes a few minutes to get used to and then becomes very convenient.

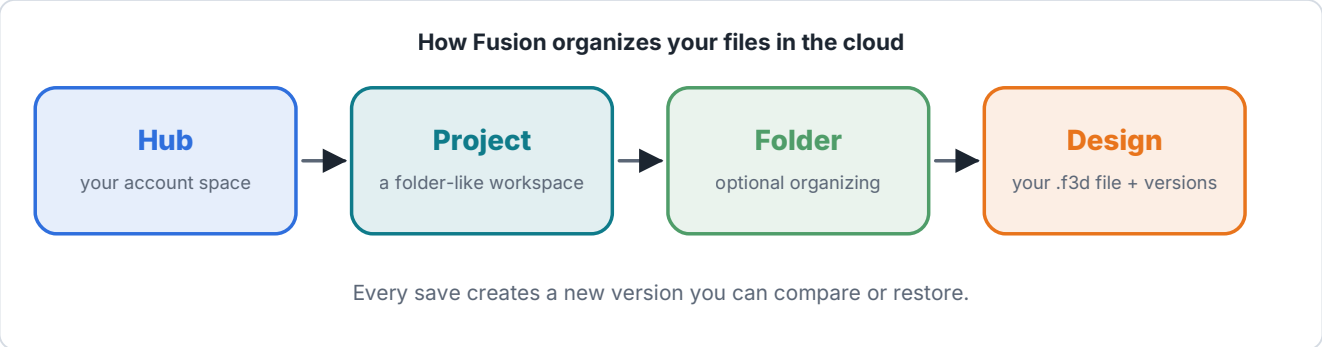


Figure 4. The cloud hierarchy: a hub contains projects, projects contain folders, and folders contain designs. Each design keeps a version history.

- **Hub**: your top-level workspace tied to your account. Hobbyists normally have one.
- **Project**: like a top-level folder. Make one project per big interest (for example "3D Printing" or "Woodworking Jigs").
- **Folder**: optional subfolders inside a project.
- **Design**: the file you work on. Fusion's native format is `.f3d`. You can also export copies to formats such as STEP, STL, 3MF and others (subject to the personal-use limits).

Open the **Data Panel** from the grid icon at the top-left of the app bar. From it you can browse projects, double-click a design to open it, see a thumbnail and the **version list**, and promote an earlier version to be the latest.

Your first design in five minutes

1. Click **File > New Design**. A blank canvas appears with the three origin planes.
2. Click **Create Sketch** on the toolbar, then click the floor plane. Fusion turns the view to look straight at it.
3. Pick **Rectangle > Center Rectangle** and draw a rectangle outward from the origin. Type `60`, press Tab, type `40`, press Enter.
4. Click **Finish Sketch** (the green check at the top-right of the toolbar).
5. Press `E` (Extrude), click inside the rectangle, drag the arrow or type `10`, then press Enter.
6. Choose **File > Save**, give the design a name, pick a project, and click Save.

Congratulations: you have made a 60 × 40 × 10 mm block and saved it with a version history. The remaining chapters explain what else you can do at every step.

> **TIP**
>
> **Set your units first.** In the Browser, expand **Document Settings** and check the units. Most hobby work (3D printing, CNC) uses millimeters. You can change the default under your profile preferences so every new design starts in the right unit.

Saving, versions and recovery

Each time you save, Fusion uploads a new version. In the Data Panel you can see the list, open any older version, and make it the current one. Because versions live in the cloud, a crashed laptop does not lose your design. It is still wise to save often with a short description such as "added lid".

CHAPTER 4

A tour of the interface

The Fusion window looks busy at first, but it has only nine parts. Learn these names and every tutorial you watch will make more sense.

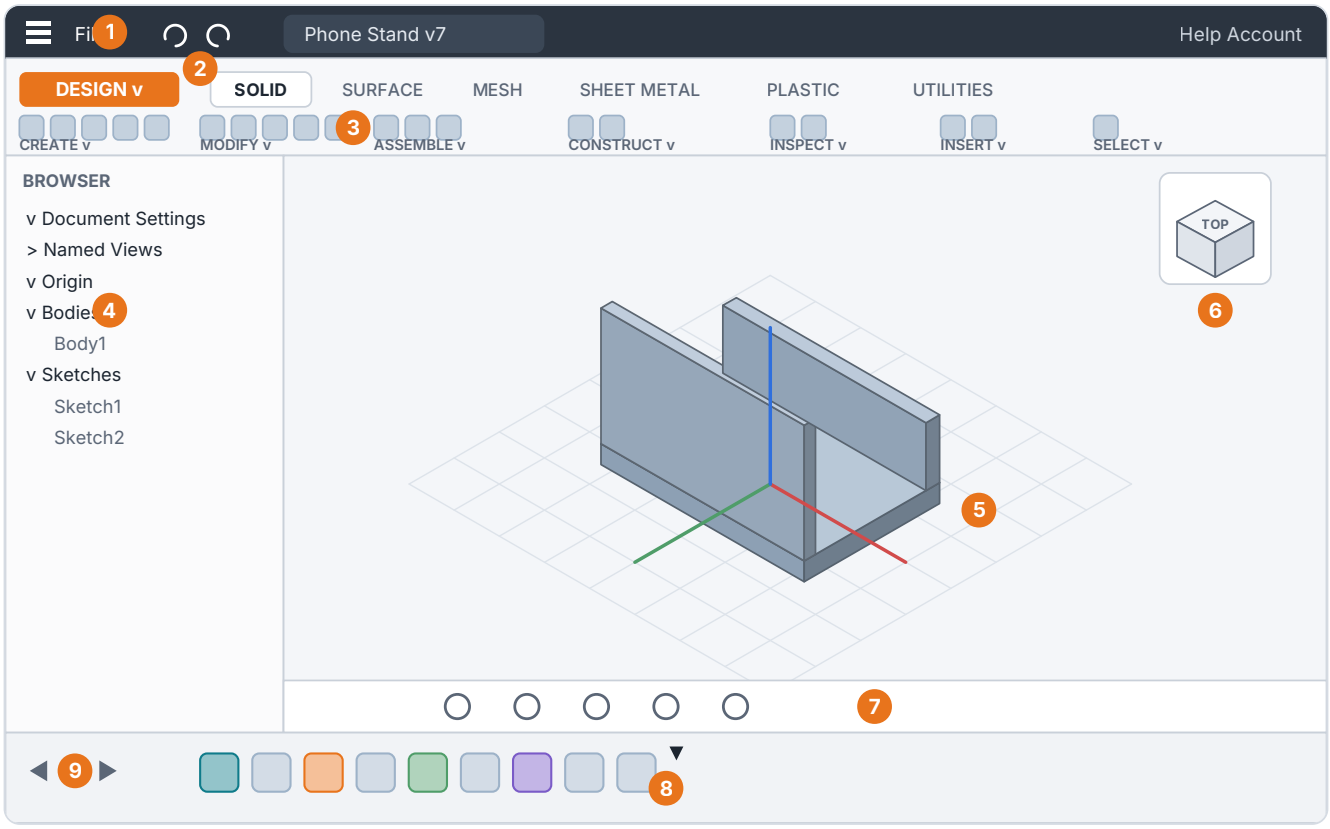


Figure 5. A schematic of the Design workspace. This is an original diagram, not a screenshot, so your version may look slightly different. Numbers match the descriptions below.

#	Part	What it does
1	Application bar	Top strip. Holds the Data Panel button, File menu, Save, Undo and Redo, open-document tabs, notifications, help and your account.
2	Workspace selector	The orange drop-down on the toolbar. Switch between Design, Render, Animation, Simulation, Manufacture, Drawing and so on.
3	Toolbar (tabs and panels)	Tools are grouped into tabs (Solid, Surface, Mesh, Sheet Metal, Plastic, Utilities) and then into panels (Create, Modify, Assemble, Configure, Construct, Inspect, Insert, Select). Click a panel's drop-down arrow to see every tool in it.
4	Browser	The tree on the left. Lists Document Settings, Named Views, Origin, Bodies, Sketches, Components and more. Use the light-bulb icons to show or hide things.
5	Canvas	The 3D view where your model lives, with a ground grid and the three origin planes.

#	Part	What it does
6	ViewCube	The little cube at the top-right for snapping to standard views.
7	Navigation bar	Bottom strip with Orbit, Pan, Zoom, Fit and the Display Settings menu.
8	Timeline	Bottom bar showing each operation as an icon, in order. Click an icon to edit that step, drag the rollback bar to travel back in time.
9	Playback controls	Step through or animate the timeline.

The toolbar in more detail

When you are in the Design workspace, the toolbar tabs are organized by kind of work:

Tab	Purpose
Solid	Everyday solid modeling and assembling. Where you will spend most of your time.
Surface	Sheet-like surfaces for complex or organic shapes; later stitched to solids.
Mesh	Editing imported STL / OBJ triangle meshes (scans, downloaded models).
Sheet Metal	Folded metal parts that unfold to a flat pattern.
Plastic	Enclosure features such as bosses, snap fits and ribs for molded or printed parts.
Utilities	Helpful extras, including 3D Print and Scripts and Add-Ins.

Within the **Solid** tab, the panels are: **Create** (make new geometry), **Modify** (change existing geometry), **Assemble** (components and joints), **Configure** (design variations, a paid feature), **Construct** (planes, axes and points), **Inspect** (measure and analyze), **Insert** (bring in outside content) and **Select** (control what you can click on). Paid plans also show an **Automate** panel. Chapters 13 to 23 cover them one by one.

Some tabs appear only when needed. When you create a sketch, a green **Sketch** contextual tab replaces the usual tabs and shows sketch tools. When you leave the sketch it disappears. Autodesk calls these *contextual tabs*.

The Browser

The Browser lists everything in your design as a tree. Understanding it is the single best way to keep a design tidy.

- **Document Settings**: units and which axis is "up" (Y-up or Z-up). Hobby 3D printing is easiest with Z-up.
- **Named Views**: saved camera positions such as Home, Top, Front.
- **Origin**: the three planes (XY, XZ, YZ), three axes and the origin point.
- **Bodies**: each solid or surface body.

- **Sketches**: all sketches, which you can show and hide. Double-click one to edit it.
- **Components**: the building blocks of assemblies, which Chapter 7 explains.

The light-bulb icon beside each item toggles visibility. Right-clicking an item opens a menu with options such as Rename, Delete, Edit Sketch, Appearance and Isolate.

The Timeline

The Timeline sits along the bottom edge. Every operation you do appears as an icon, in order, from left to right. This is the feature that makes Fusion **parametric**.

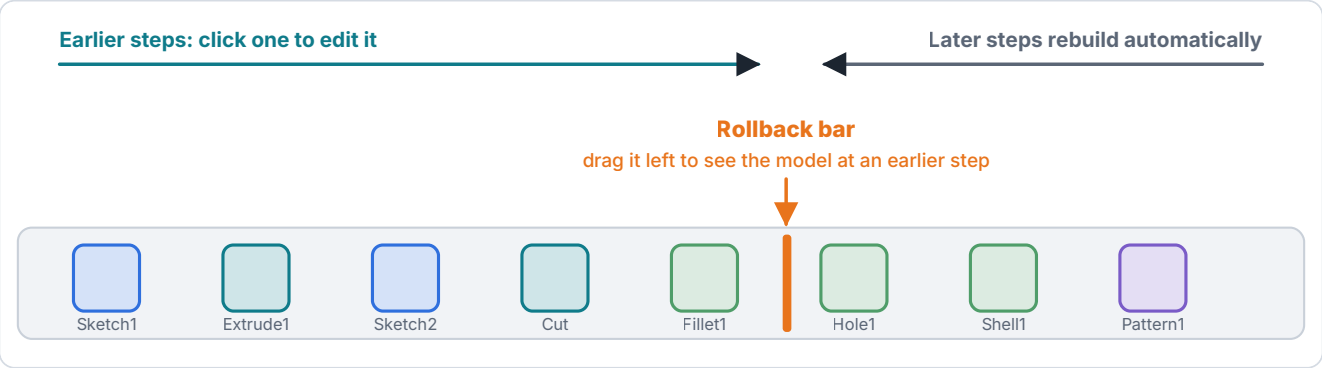


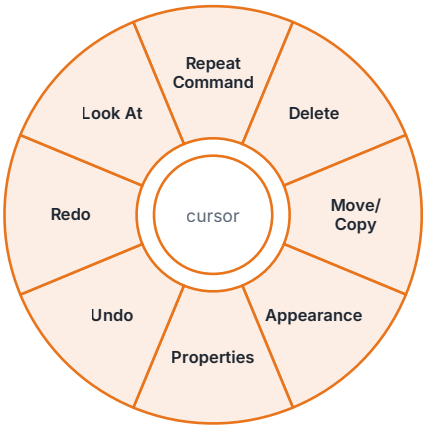
Figure 6. The timeline records each step. Drag the rollback bar to see earlier states, or double-click an icon to edit that step; later steps rebuild automatically.

- **Edit a step**: double-click its icon, change a value, press OK. Everything after it updates.
- **Roll back**: drag the black rollback marker left. Anything you add now is inserted at that point in history.
- **Reorder**: drag a feature left or right. Some moves break later steps and show a red warning.
- **Group**: select several icons, right-click and choose Create Group to keep a long timeline readable.
- **Playback**: the controls at the left of the timeline replay your model being built, like a time-lapse.

> **TIP**
> Name your features. Right-click any timeline item and choose **Rename**. "Cable slot" is far more useful than "Extrude14" when you return to a design after a month.

The Marking Menu and shortcuts

Right-click on the canvas and a circular **marking menu** appears around your cursor, with the most useful commands for what you clicked, plus a list of more options beneath it. Flick the mouse toward a wedge to choose it without aiming carefully; with practice this is very quick.



The Marking Menu

Right-click (or press & hold the right mouse button) on the canvas.

Flick toward a wedge for a fast gesture: no need to aim at tiny buttons.

Below the wheel, a list shows more options for the thing you clicked.

Figure 7. The Marking Menu offers fast access to common commands. Exact wedges vary with what you right-click.

Press `S` to open a quick-access palette where you can type the name of almost any command and pick it from the results. This is the fastest way to find a tool you cannot locate on the toolbar.

CHAPTER 5

Navigating the 3D canvas

Moving around a 3D model is the first skill to master. Fusion's defaults suit a three-button mouse; a trackpad or SpaceMouse also works.

Mouse controls

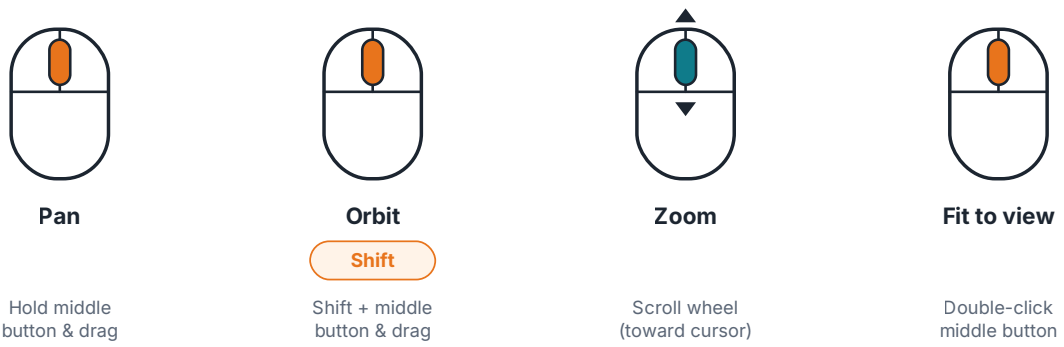
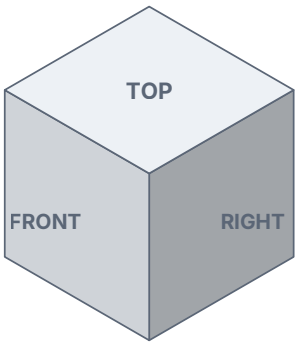


Figure 8. The default Fusion navigation with a three-button mouse. A cheap mouse with a clickable scroll wheel is the best hobbyist investment you can make.

Action	Default control
Pan (slide the view)	Hold the middle mouse button and drag
Orbit (rotate)	Hold `Shift` + middle mouse button and drag
Zoom	Scroll the wheel. Fusion zooms toward your cursor by default, which is a setting you can change.
Fit everything	Double-click the middle mouse button
Select	Left-click. Use `Ctrl` (Windows) or `Cmd` (Mac) to add to the selection.
Marking menu	Right-click
Cancel a command	`Esc`

If you prefer other mouse mappings, open your profile name > **Preferences > General > Default Navigation Mode**. Fusion lets you switch the mapping to match programs such as Alias, Inventor, Tinkercad or SolidWorks, so existing muscle memory still works. No mouse? Use the Orbit, Pan and Zoom buttons on the navigation bar.

The ViewCube



- Click a face → snap to that standard view
- Click an edge → view from that edge (45°)
- Click a corner → isometric view
- Drag the cube → orbit freely
- Click the little house → return to Home view

Tip
Right-click the cube → “Orient to...”, “Perspective”, “Orthographic” options.

Figure 9. The ViewCube at the top-right of the canvas. Clicking its faces, edges and corners gives you standard orthographic and isometric views.

Display settings and visual styles

The navigation bar's **Display Settings** menu controls how the model is drawn. Useful options include:

- **Visual Style**: Shaded, Shaded with Visible Edges, Shaded with Hidden Edges, Wireframe, Wireframe with Hidden Edges, Wireframe with Visible Edges.
- **Environment and Effects**: ground shadows, reflections, ambient occlusion, anti-aliasing (turning some off helps on slower computers).
- **Camera**: switch between **Orthographic** (parallel lines, best for technical work) and **Perspective** (natural looking).
- **Grid and Snaps**: show or hide the grid and set snap behavior for sketching.
- **Objects Visibility**: quickly hide the origin, sketches, construction geometry, joints and more.

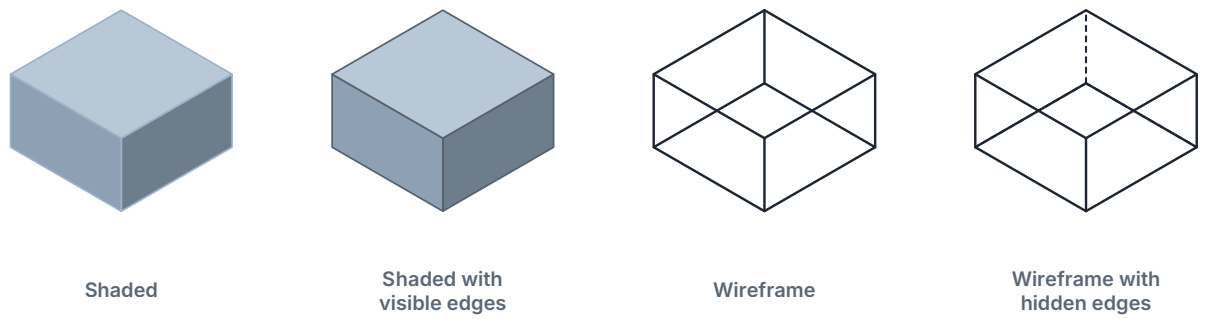


Figure 10. Four common visual styles. Wireframe with hidden edges is handy for checking holes and internal features.

Useful navigation habits

1. **Fit often.** Double-click the middle mouse button whenever you feel lost.

2. **Use the ViewCube's Home button** to reset to your design's saved Home view.
3. **Sketch in 2D.** When you start a sketch, Fusion automatically turns the view to face the sketch plane. Do not orbit while sketching unless you must.
4. **Save Named Views** for angles you reuse: right-click Named Views in the Browser, then choose New Named View.
5. **Orthographic for accuracy, perspective for pictures.** Orthographic views make it far easier to judge alignment.

2

PART 2

Thinking in Fusion

Before touching more tools, it helps to understand the handful of ideas that everything else in Fusion is built on: history-based modeling, planes and the origin, and how to organize parts into components.

Chapter 6 · Core concepts Chapter 7 · Components, bodies and the Browser Chapter 8 · Parameters, units, names and appearances

CHAPTER 6

Core concepts

Five ideas explain most of Fusion's behavior. Spend ten minutes here and the rest of the program feels logical instead of mysterious.

1. Parametric, history-based modeling

In a **parametric** program the model is a recipe, not a lump of clay. Fusion stores each operation ("sketch a rectangle 60 × 40", "extrude it 10 mm", "fillet these four edges 2 mm") in the timeline. When you change an earlier value, Fusion replays the later steps against the new geometry. That is why a 60 mm box can become a 90 mm box by editing a single number.



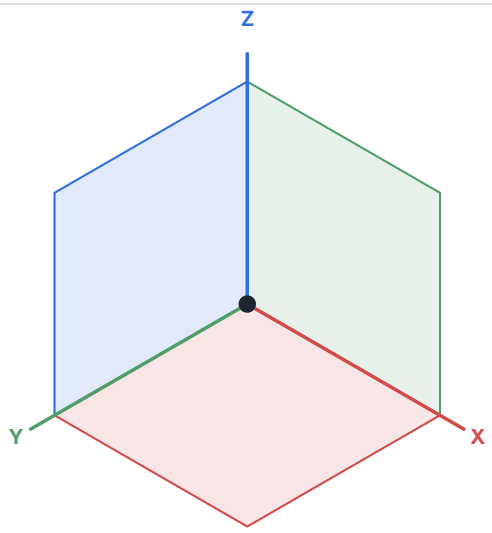
Figure 11. One parameter-driven design resized three ways. Fusion has a Parameters table (Modify > Change Parameters) where named values can drive dimensions and even each other.

KEY IDEA

Design intent. Good Fusion models record *why* a feature is the size it is. A hole that is "always centered" should be constrained to the center, not placed at a typed coordinate that happens to be the center today. Then, when the plate gets wider, the hole follows.

2. The origin and the three planes

Every design has an **origin** (the point 0, 0, 0) and three **planes** through it. Sketches are drawn on a plane or on a flat face of an existing body. Fusion lets you choose **Y-up** or **Z-up** (a preference sets the default for new designs); Z-up matches 3D printers and CNC machines, where Z is height, so many hobbyists switch it on for new designs.



XY plane ("Top" in Z-up)
Floor / table surface. Most first sketches start here.

XZ plane ("Front")
Vertical plane facing you.

YZ plane ("Right")
Vertical plane facing the side.

Origin = the point where all three planes meet (0, 0, 0).

Figure 12. Origin, axes (red = X, green = Y, blue = Z) and the three origin planes. The plane called "Top" depends on whether the document is Y-up or Z-up.

Anchoring your first sketch to the origin (for example, centering a rectangle on it) gives the design a stable reference that later features can lean on.

3. Sketch, then feature

A **sketch** is a flat drawing. A **feature** turns it into 3D: Extrude, Revolve, Sweep, Loft and so on. The sketch stays linked to the feature, so editing the sketch changes the solid.

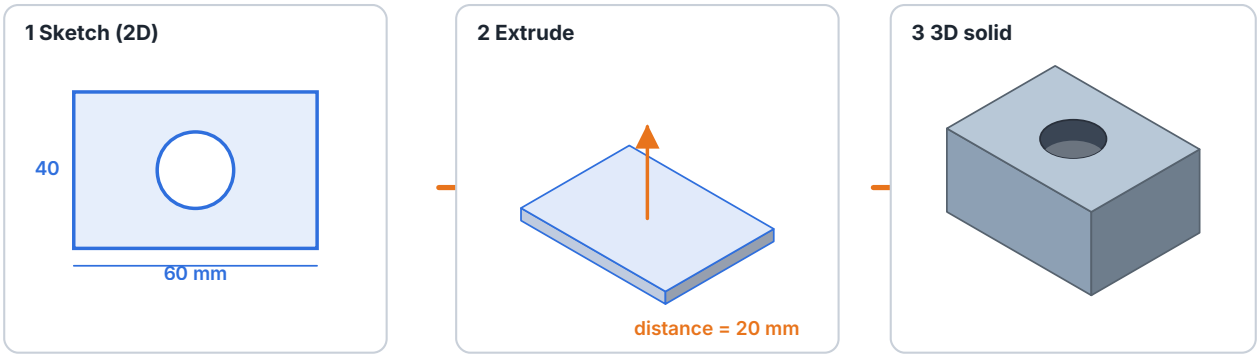


Figure 13. A circle inside a rectangle becomes, after Extrude, a block with a hole. The circle becomes a hole because Fusion treats the inner profile as a separate closed area.

Fusion highlights each closed region of a sketch as a **profile**. When you extrude, you choose which profile or profiles to use. Selecting only the outer ring or only the circle gives different results from the same sketch.

4. Bodies and components

A **body** is a single solid or surface. A **component** is a container that can hold bodies, sketches, joints and other components. Parts that move or print separately should be separate components (Chapter 7).

5. Everything is editable and nothing is final

The Undo button (`Ctrl`+`Z` or `Cmd`+`Z`) is your friend; so is the timeline, and so is cloud version history. Beginners often freeze because they fear "breaking" the model. Fusion is designed so that you can always step back.

Fusion design vocabulary at a glance

Term	Meaning
Sketch	A flat set of lines, arcs, circles and dimensions on a plane.
Profile	A closed region inside a sketch that can be extruded or revolved.
Feature	A 3D operation stored in the timeline (Extrude, Fillet, Hole…).
Body	One solid or surface shape.
Component	A named part container with its own origin and contents.
Occurrence	A copy (instance) of a component in an assembly.
Joint	A rule that locks or limits how two components can move.
Parameter	A named value (such as `wall`) used by features and sketches.
Timeline	The ordered list of features that Fusion replays when you change something.

CHAPTER 7

Components, bodies and the Browser

The most common beginner mistake in Fusion is putting everything in one component. A little organization now prevents a painful rebuild later.

Body vs component

When you extrude a sketch with the default **New Body** or **Join** operation, you create geometry inside the current component. If you create three separate parts this way, they all share the same sketches, the same origin and the same timeline space. You cannot give them joints, reposition them independently, or export them cleanly one at a time.

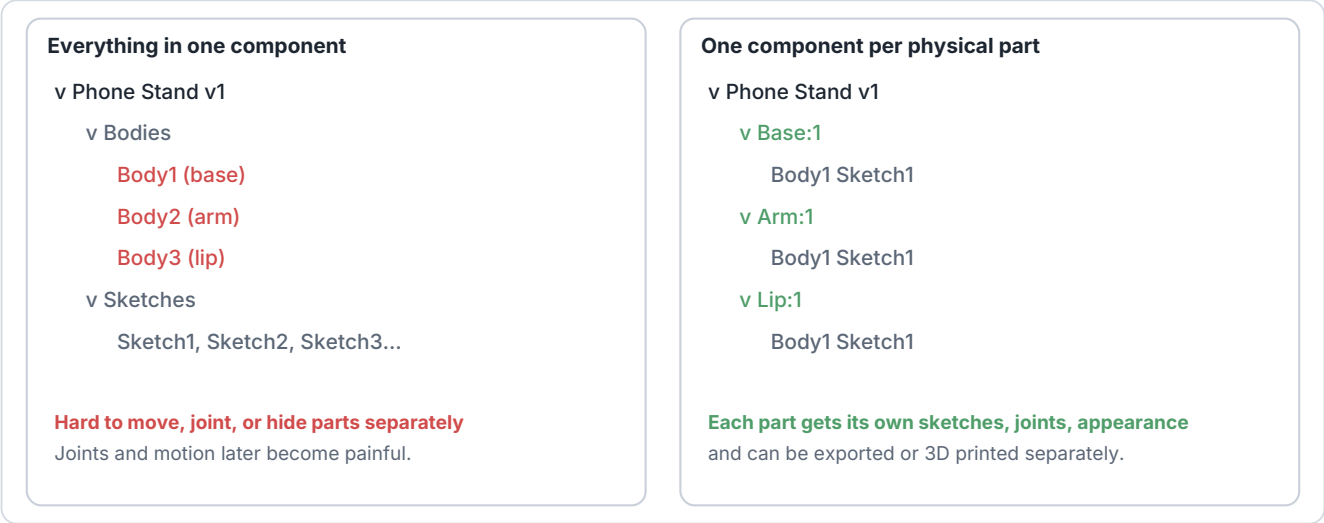


Figure 14. Left: every body lives in one component. Right: each physical part has its own component, which is the arrangement joints, motion and printing prefer.

How to use components well

1. **Decide what is a "part".** If it is a separate physical object, or will be printed or cut separately, it deserves its own component.
2. **Create the component first.** In the Browser, right-click the top-level item and choose **New Component**, name it, and make sure it is **activated** (the radio button beside it).
3. **Sketch and extrude inside the active component.** Fusion places new sketches and bodies in whichever component is active.
4. **Use Extrude's "New Component" operation** when you want an extruded shape to become its own part instantly.
5. **Ground one component.** Right-click the base part and choose **Ground** (or rely on the first joint) so the assembly has a fixed reference.

WATCH OUT

"New Component" in the Create panel makes an *empty* component. If you then draw while a different component is still active, your geometry lands in the wrong place. Always check which component has the filled activation dot before you sketch.

Top-down vs bottom-up

Approach	How it works	Best for
Top-down	Design everything in one file, creating components as you go, so parts are sized relative to each other.	Enclosures, lids, things that must fit together.
Bottom-up	Design each part in its own file, then use **Insert > Insert into Current Design** to assemble them.	Reusable parts such as a standard bracket or knob.

Hobbyists usually start with top-down because it avoids managing many files.

Useful Browser actions

Action	How
Show / hide	Click the light bulb beside any item.
Isolate	Right-click > Isolate to show only that item. Use Show All to return.
Rename	Slow double-click the name or press F2.
Activate component	Click the radio button beside it.
Change appearance	Right-click > Appearance (or press `A`) and drag a material onto the body.
Find in Timeline	Right-click > Find in Timeline to jump to the feature that made it.

CHAPTER 8

Parameters, units, names and appearances

Four small habits that make designs easier to change, share and print.

Units and precision

Check **Document Settings > Units** when you start. A 3D-print slicer expects millimeters. If you ever import a model that arrives 25.4 times too big or too small, it is almost always an inch/millimeter mix-up. You can type units into any field (`2 in`, `5 cm`) and Fusion converts them.

The Parameters table

Open **Modify > Change Parameters**. The table has two sections:

- **User Parameters** that you create, such as `width`, `wall` or `screw_d`, with a value or expression (for example `width / 10`).
- **Model Parameters**, which Fusion creates automatically for every dimension and feature value (`d1`, `d2`, ...).

You can type a parameter name into any numeric field, so five dimensions can all depend on `wall`. Change `wall` from 2 to 2.4 and everything updates. Parameters can be favorited so they appear at the top, and they can include units and comments.

> **TIP**
>
> Hobbyist habit: create parameters for **tolerances** such as `clearance = 0.3 mm`. When your printer or filament changes, adjust one number and re-export.

Naming

Rename components, bodies and key features early. Short, plain names like "Lid" and "Hinge pin" beat "Component12". It makes timelines readable and helps when exporting several parts at once.

Appearances and physical materials

Two different things share the "material" idea:

- **Appearance** is how something looks: color, finish, texture. Press `A` to open the Appearance panel and drag a swatch onto a body, face or component.
- **Physical Material** contains real properties such as density and strength. It lets Fusion compute mass and drives simulation. Right-click a body > Physical Material, or use Modify > Physical Material.

A physical material of PLA, for example, lets the Properties dialog report the weight of a part, which helps you estimate filament use.

Document settings worth knowing

Setting	What it affects
Units	The default unit for typing values.
Model Up Orientation	Y-up or Z-up. Change it before modeling, not after.
Capture Design History	On = parametric timeline. Off = direct modeling with no history (Chapter 24).
Named Views	Saved camera positions you can return to.

3

PART 3

Sketching

Every solid in Fusion starts with a sketch. This part covers the sketch workflow, all of the drawing and editing tools, and the constraint and dimension system that makes sketches precise and easy to change.

Chapter 9 · Sketch basics Chapter 10 · Sketch Create tools Chapter 11 · Sketch Modify tools Chapter 12 · Constraints and dimensions

CHAPTER 9

Sketch basics

A sketch is a flat drawing on a plane. Learn to start, finish and tidy sketches and the rest of modeling becomes straightforward.

Starting a sketch

1. On the **Solid** tab, click **Create Sketch** (the first icon in the Create panel). You can also press nothing at all: choosing **Sketch** from the Create menu is the same command.
2. Click a plane (XY, XZ or YZ) or a **flat face** of an existing body. Fusion rotates the view to face it and shows the sketch tools.
3. Draw with sketch tools, add dimensions and constraints.
4. Click **Finish Sketch** at the right end of the toolbar.

The Sketch contextual tab

While sketching, the toolbar shows a dedicated **Sketch** tab with these panels:

Panel	What is in it
Create	Line, Rectangle, Circle, Arc, Polygon, Ellipse, Slot, Spline, Conic Curve, Point, Text, Mirror, Circular Pattern, Rectangular Pattern, Project / Include, Sketch Dimension.
Modify	Fillet, Trim, Extend, Break, Sketch Scale, Offset, Move/Copy, Delete, Change Parameters.
Constraints	Coincident, Collinear, Concentric, Midpoint, Fix/Unfix, Parallel, Perpendicular, Horizontal/Vertical, Tangent, Smooth, Symmetry, Equal.
Inspect	Measure, Interference and other analysis tools that work inside a sketch.
Insert	Insert image (canvas), DXF, SVG and other drawing data.
Select	Selection filters.

The Sketch Palette

When you are in a sketch, a small **Sketch Palette** appears at the right. It holds options that change how you work:

Option	Effect
Linetype	Normal vs **Construction** vs Centerline (shortcut `X` toggles construction). Construction lines guide the design but are ignored when you extrude.
Look At	Turns the view square to the sketch plane.

Option	Effect
Sketch Grid / Snap	Shows the grid and snaps the cursor to it.
Slice	Cuts away the bodies in front of the sketch plane so you can see what you are working on.
Show Profile	Shades closed profiles so you can check the shape is truly closed.
Show Points	Shows all sketch points.
Show Dimensions / Constraints	Hide or reveal the dimension and constraint icons.
Show Projected Geometries	Reveals geometry you projected from other bodies.
3D Sketch	Lets you draw lines in 3D space instead of on one plane.

Drawing with typed values

Most sketch tools let you type exact values while drawing. After the first click, a small input box appears. Type a number, press `Tab` to move to the next box (for a rectangle: width, then height), and press `Enter` to lock it in. That one habit produces fully dimensioned sketches with almost no extra work.

Sketch on a face

To add a hole or boss to an existing body, sketch on its flat face. Fusion links the sketch to that face. If you later move or resize the face, the sketch stays attached (it is "parametrically attached"). Use **Project / Include** (Chapter 10) to bring edges of the body into the sketch as references.

Editing a sketch later

Double-click the sketch in the Browser or its timeline icon, or right-click > **Edit Sketch**. Changes flow into all features built from it.

> **WATCH OUT**
>
> If you finish a sketch and start another one on the same plane, Fusion does **not** connect the two. Lines from different sketches may look joined but are separate, which is a common reason an extrude will not "see" a closed shape. Keep related geometry in one sketch, or use Project / Include.

CHAPTER 10

Sketch Create tools

All the drawing tools in the Sketch Create panel, with what each does and where hobbyists use it.

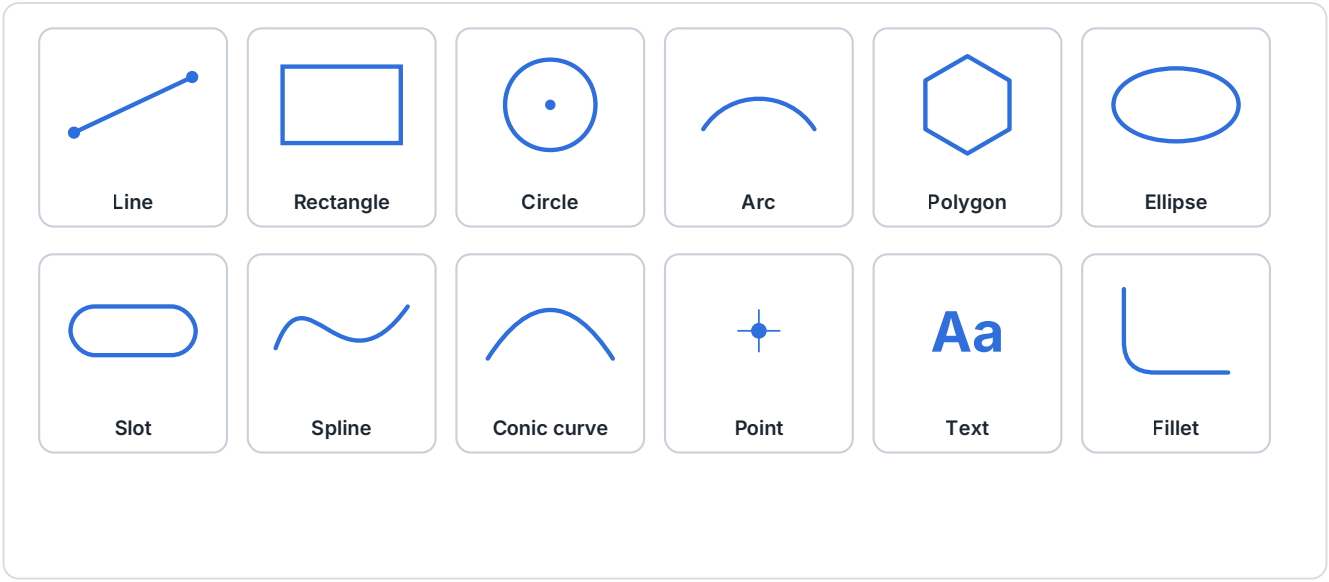


Figure 15. Core sketch tools. Each has options inside its drop-down in the Create panel.

Lines, rectangles and circles

Tool	What it does	Key
Line	Click start and end points. Keep clicking for connected segments; right-click or `Esc` finishes. Drag from the end point to switch to a tangent arc.	`L`
2-Point Rectangle	Corner to opposite corner.	`R`
3-Point Rectangle	Two points set one side and angle, third sets width. Good for tilted rectangles.	
Center Rectangle	Draws from the center outward.	
Center Diameter Circle	Click the center, then drag or type a *diameter*.	`C`
2-Point / 3-Point Circle	Circle through two ends of a diameter or three points on the edge.	
2-Tangent / 3-Tangent Circle	Circle that touches two or three lines or arcs. Excellent for rounding between shapes.	

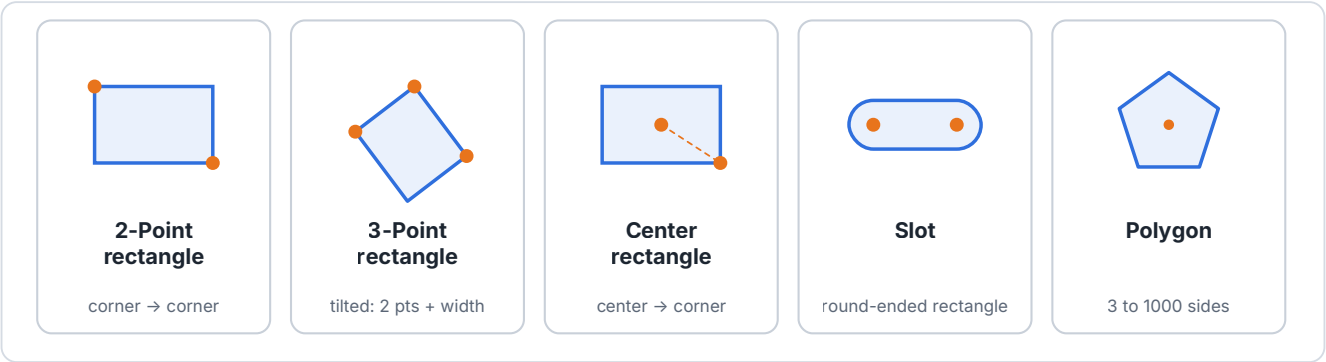


Figure 16. Rectangle, slot and polygon variants. Orange dots show where you click.

Arcs, polygons, ellipses and slots

Tool	What it does
3-Point Arc	Start, end, then a point the arc passes through.
Tangent Arc	Starts from the end of a line or arc and continues smoothly.
Center Point Arc	Click center, start, then end angle. Best for exact radii.
Polygon	Circumscribed, inscribed or edge polygon with a chosen number of sides. Use it for hex nuts, star knobs and honeycomb patterns.
Ellipse	Center plus two axes.
Slot	Center-to-center, overall, center-point arc or 3-point arc slot. The perfect shape for adjustable screw slots.

Curves, points and text

Tool	What it does
Fit Point Spline	A smooth curve that passes *through* every point you click. Intuitive for organic outlines.
Control Point Spline	A curve shaped by a control polygon; points pull the curve toward them. Smoother and easier to fine-tune.
Conic Curve	Smooth parabola-like arc between two endpoints, shaped by a "rho" value. Great for gentle design curves.
Point	A sketch point, useful as a reference for holes, patterns and splines.
Text	Adds editable text. Choose font, height and alignment. The text becomes profiles you can extrude or cut (for labels and name plates). Text can also follow a path.

TIP

Autodesk's April 2026 update notes added **Text on Path** support for multiple text instances along one path and a smarter Mirror that auto-selects a single valid centerline, which are both small time-savers for hobby projects.

Mirror and patterns in a sketch

- **Mirror**: pick geometry, then a mirror line (usually a centerline). Draw half of a symmetric shape and mirror the rest.
- **Circular Pattern**: repeats sketch geometry around a center point. Use for bolt circles.
- **Rectangular Pattern**: repeats geometry along two directions with set spacing. Use for grids of holes.

Project / Include

This panel brings reference geometry into the current sketch:

Command	Use
Project	Projects edges, faces or other sketch curves onto the sketch plane. The result stays linked to the source.
Intersect	Finds where a body or face crosses the sketch plane.
Include 3D Geometry	Brings 3D edges in directly for use in a 3D sketch or path.
Intersection Curve	Curve where two bodies meet.
Project to Surface	Wraps sketch geometry onto a curved face.

HOBBYIST ANGLE

Project is how you make a lid that matches a box: sketch on the lid plane, project the box outline, offset it inward by your clearance, and extrude. When the box changes, so does the lid.

CHAPTER 11

Sketch Modify tools

After drawing rough geometry, these tools clean it up.

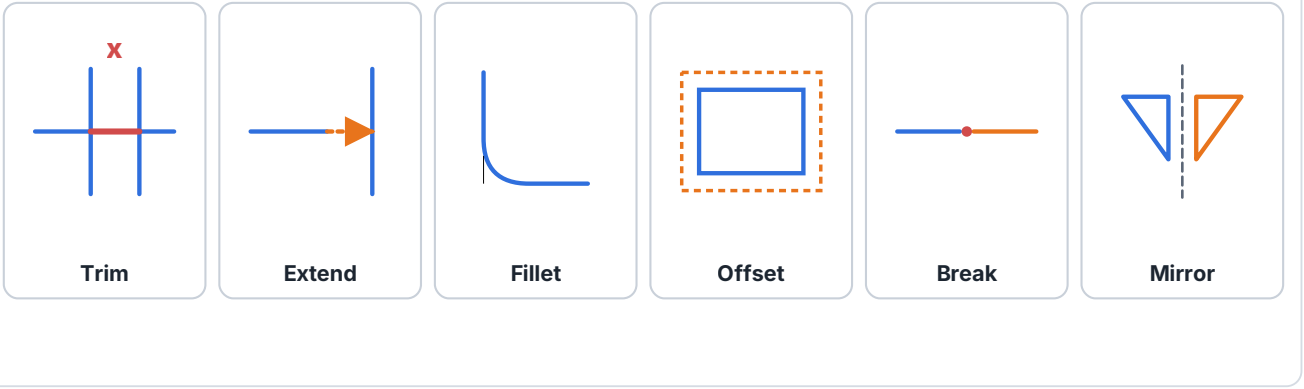


Figure 17. Common sketch modification tools.

Tool	What it does	Key
Fillet	Rounds a sharp corner between two lines or arcs with a typed radius.	
Trim	Deletes the segment of a line or curve between its nearest intersections. The fastest way to clean overlaps.	`T`
Extend	Lengthens a line or curve until it hits the next object.	
Break	Splits a curve at its intersections with other geometry, without deleting anything, so each part can be edited or dimensioned separately.	
Sketch Scale	Scales selected geometry from a base point by a factor. Handy for traced images.	
Offset	Creates a parallel copy at a set distance. Offsets entire chains of connected curves, perfect for wall thickness outlines and clearances.	`O`
Move / Copy	Moves or copies sketch entities by distance or angle.	`M`
Stretch	Moves a portion while stretching connected geometry.	
Delete	Removes selected geometry (also the `Delete` key).	
Change Parameters	Opens the Parameters table without leaving the sketch.	

A clean-up routine that works

1. Draw the rough outline with lines and arcs, not worrying about precision.
2. Use **Trim** to remove overlaps and extras.

3. Add **Fillet** to corners that should be rounded.
4. Add constraints, then dimensions (Chapter 12) until the geometry turns black.
5. Toggle **Show Profile** to confirm the outline is closed before finishing.

> **TIP**
> Sketch fillets are fine, but 3D fillets (Chapter 17) are usually better for rounding finished corners: they can be changed after the shape is made, and they can stay attached to edges if the shape changes.

CHAPTER 12

Constraints and dimensions

Constraints are rules ("these two lines are parallel") and dimensions are measurements ("this line is 40 mm"). Together they turn a rough drawing into a precise, predictable one.

The colors tell you the state

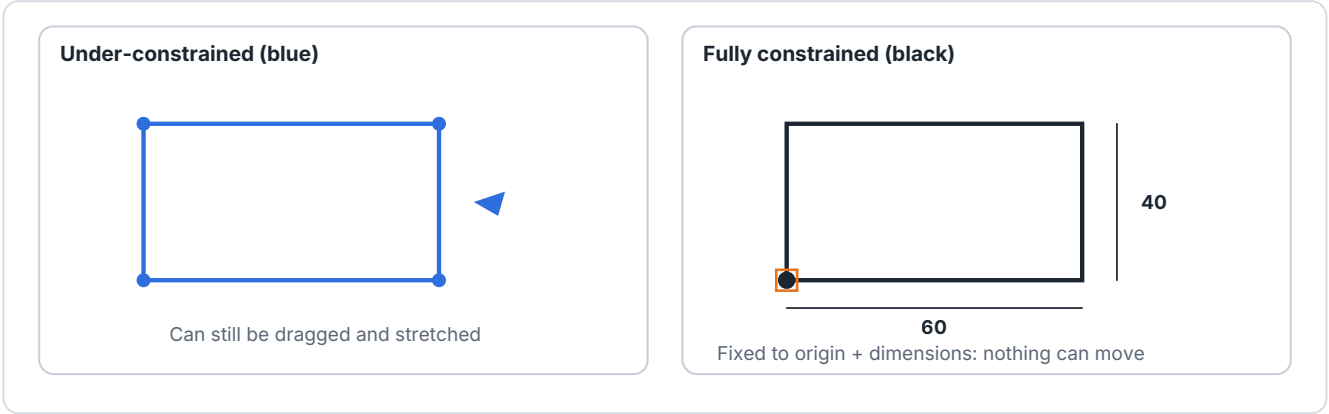


Figure 18. Blue geometry can still move because it is under-constrained. Black geometry has no remaining freedom.

A sketch that is fully constrained is more predictable and easier to modify. Autodesk's own guidance is to aim for that, but it is not strictly required: plenty of casual hobby sketches work fine while partly blue. Be strict when the part must fit something else.

Geometric constraints

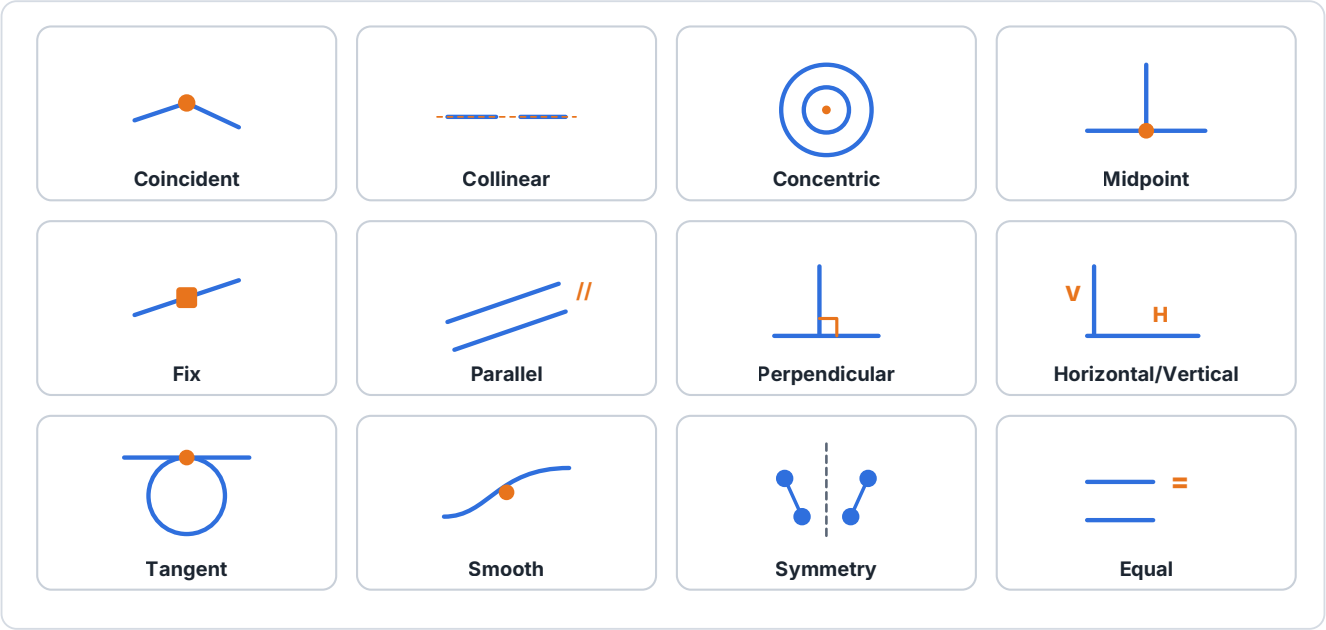


Figure 19. The twelve sketch constraints in Fusion.

Constraint	Meaning
Coincident	Two points meet, or a point lies on a curve.
Collinear	Two lines lie along the same infinite line.
Concentric	Two circles or arcs share a center.
Midpoint	A point sits at the middle of a line or arc.
Fix / Unfix	Locks geometry in place so nothing can move it.
Parallel	Two lines run in the same direction.
Perpendicular	Two lines meet at 90°.
Horizontal / Vertical	A line is parallel to the X (or Y) axis; or two points line up horizontally or vertically.
Tangent	A line and an arc (or two arcs) touch smoothly at one point.
Smooth	A spline meets a line or arc with a smooth, curvature-continuous join.
Symmetry	Two objects mirror each other across a line.
Equal	Two lines have equal length, or two circles or arcs have equal radius.

Fusion also adds many constraints **automatically** as you draw. If your cursor shows a small glyph next to it (for example a horizontal line mark), a constraint is about to be applied. Hold `Ctrl` while drawing to turn off auto-snaps when they get in the way.

Sketch Dimension (the D key)

Press `D`, click a line (or two points, or a circle), move the mouse to place the dimension, then type the value and press Enter. Fusion adapts the type automatically:

- One line → length. Two lines → angle. Circle → diameter. Arc → radius.
- A point and a line → perpendicular distance.
- Right-click while placing for options such as Aligned, Horizontal and Vertical.

A dimension can use an expression or a parameter name (for example `width/2`). **Driven (reference) dimensions** show a measurement without controlling the geometry: right-click a dimension and choose "Driven Dimension" to make it a display-only value.

Fixing common sketch problems

Problem	Likely cause and fix
Extrude will not select a profile	The outline is not closed. Use Show Profile; look for a tiny gap, and drag endpoints together (a Coincident constraint should appear).
"Over-constrained" warning	A dimension conflicts with an existing constraint. Delete the conflicting one or make it driven.
A sketch collapsed when I added a dimension	Conflicting rules or a wrong value. Undo, and dimension the longest, most important edges first.
Geometry keeps jumping when I drag it	It is under-constrained. Add dimensions, or Fix one anchor point to the origin.
Tiny line segments appear	Fast mouse movement. Use Trim or Delete to remove them. April 2026 notes say Fusion now filters unintended lines caused by sensitive mouse movement.

KEY IDEA

A reliable order of operations: (1) draw roughly, (2) add Coincident, Horizontal/Vertical, Tangent, Equal and Symmetry constraints, (3) fix one point to the origin, (4) add dimensions starting with overall size, (5) finish when the lines turn black.

4

PART 4

Solid modeling

This is the heart of Fusion for most hobbyists. Every tool in the Solid tab is covered: Create, Modify, Construct, Inspect, Insert and Select, with pictures, key options, and advice on when to use each.

Chapter 13 · The Create menu 14 · Extrude 15 · Revolve, Sweep, Loft and friends 16 · Holes and threads 17 · The Modify menu 18 · Fillet, Chamfer, Shell and Draft 19 · Patterns and Mirror 20 · Construct 21 · Inspect 22 · Insert 23 · Select

CHAPTER 13

The Create menu

The Create panel is where new geometry is born. Here is the complete list at a glance, then deeper chapters for the big tools.

Create panel overview

Tool	What it does	Good for
New Component	Creates an empty component (part container).	Organizing parts
Create Sketch	Starts a 2D sketch on a plane or face.	Everything
Extrude E	Pushes a profile straight into 3D, adding or cutting material.	Plates, blocks, pockets
Revolve	Spins a profile around an axis.	Knobs, cups, wheels
Sweep	Drags a profile along a path.	Handles, pipes, rails
Loft	Blends smoothly between two or more profiles.	Bottles, transitions
Rib	Adds a thin support wall between a sketch curve and a body.	Brackets, reinforcement
Web	Adds a network of ribs inside a hollow body.	Stiffening shells
Emboss	Raises or engraves sketch shapes or text onto curved or flat faces.	Logos, labels
Hole H	Simple, counterbore or countersink holes, with optional threads.	Screws, bolts
Thread	Adds cosmetic or modeled threads to cylinders or holes.	Bolts, bottle caps
Box, Cylinder, Sphere, Torus	Primitive solids made in one dialog.	Fast starting shapes
Coil	Helical spring or thread shape.	Springs, screw threads
Pipe	Hollow or solid tube along a path.	Tubes, cables, wire frames
Pattern	Rectangular, Circular or Path pattern of features, bodies or components.	Repeats (Chapter 19)
Mirror	Duplicates across a plane.	Symmetrical parts
Thicken	Turns a surface into a solid with thickness.	Surface workflows

Tool	What it does	Good for
Boundary Fill	Creates a solid from the space enclosed by tool bodies or surfaces.	Closed shapes from surfaces
Create Form	Enters the T-Spline (Form) environment.	Organic shapes (Chapter 26)
Create Base Feature	Lets you edit imported or direct-modeled geometry without recording history.	Imported models
Derive	Brings geometry from another design into this one, linked to the source.	Multi-file projects
Create PCB	Starts an electronics PCB from an outline.	Circuit boards

Primitives: the quick starts

Box, **Cylinder**, **Sphere** and **Torus** each open one dialog. Pick a plane, click to place the center, and type dimensions. They are convenient for prototypes, but a sketch-based shape is usually better for finished parts because you can constrain and dimension it later. Primitives also create their own sketch, so you can still edit them through the timeline.

Which tool should I use?

If your part looks like...	Start with
A flat plate with holes or cut-outs	Sketch + Extrude
A shape that is round when viewed from the top (cup, knob, wheel)	Sketch profile + Revolve
A bent rod, handle or rail	Sketch path + profile + Sweep
A transition from one shape to another	Two sketches + Loft
A hollow box	Extrude, then Shell
A coiled spring or a thread	Coil
A smooth organic form (mouse, character)	Create Form

CHAPTER 14

Extrude

Extrude is the workhorse of solid modeling, and more than half of hobby parts need nothing else. Learn every option here and you will model quickly.

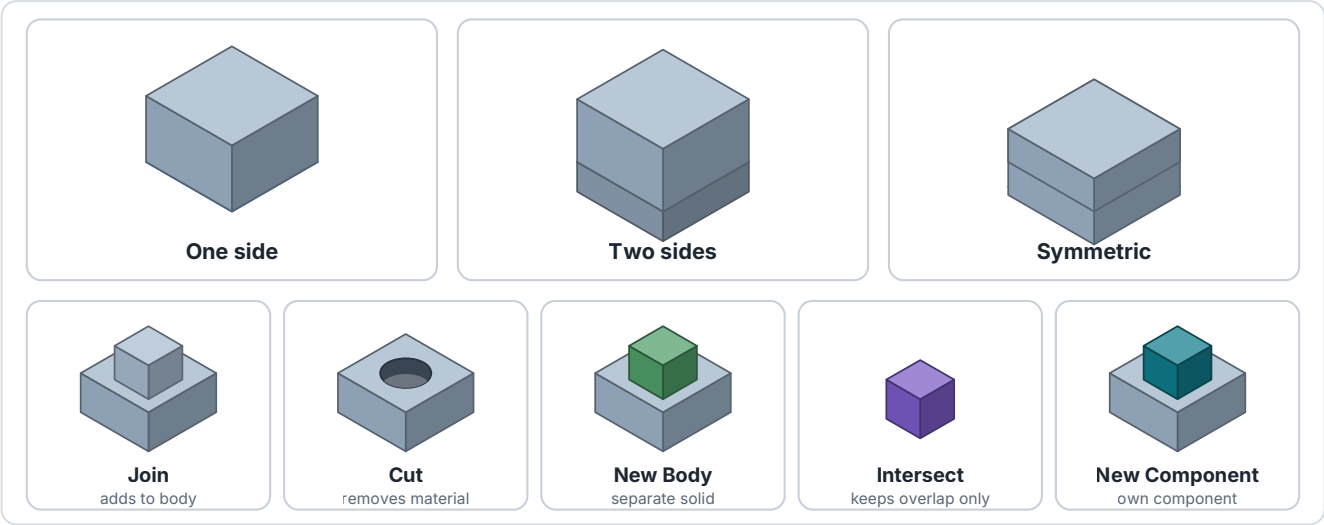


Figure 20. Top: the three direction types. Bottom: the five operations. Choosing the right operation decides whether you add, remove or separate material.

The Extrude dialog, field by field

Field	Meaning
Profile	Click one or more closed regions of a sketch. Hold `Ctrl` / `Cmd` (or Shift) to add more.
Start	Where the extrusion begins: from the Profile Plane (default), an **Offset** plane, or an **Object** (a face or plane).
Direction	**One Side**, **Two Sides** (a different distance each way) or **Symmetric** (equal both ways, a good way to center a part).
Extent type	**Distance** (type a number), **To Object** (stop at a face, body or plane) or **All** (through everything).
Distance	The length. Drag the arrow handle or type. Negative values reverse the direction.
Taper Angle	Tilts the sides inward or outward (see below).
Operation	**Join** merges with an existing body. **Cut** removes material. **Intersect** keeps only the overlap. **New Body** creates a separate solid. **New Component** creates a separate component.

Field	Meaning
Thin Extrude	For open or closed profiles, produces a wall of set thickness rather than a solid slab. Great for tubes and brackets.

Operations explained

- **Join** adds the new shape to the body it touches. Use it to build up a base with bosses and ribs.
- **Cut** subtracts. Sketch a shape on a face, Extrude with Cut and the shape becomes a pocket. With **Extent = All** it goes completely through.
- **New Body** is how you create a second solid that should not merge, like a lid sitting above a box.
- **Intersect** is less common but excellent for making a shape from two orthogonal outlines (the front and side silhouettes).
- **New Component** goes one step further than New Body by making the result its own component.

Taper

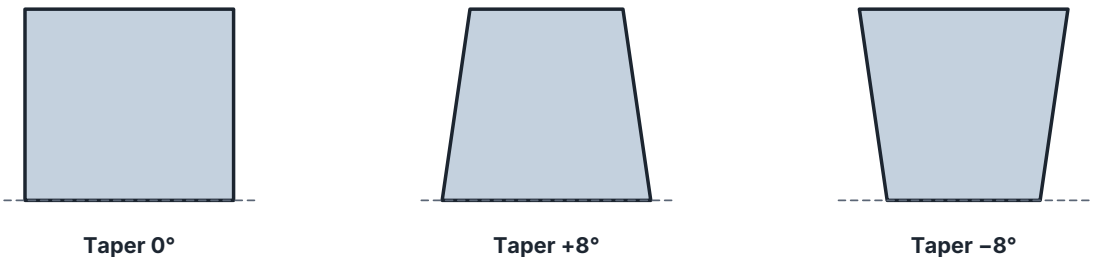


Figure 21. Taper angle. A small positive taper on pockets and walls helps parts release from molds and can make printed walls cleaner.

Extrude from selected faces

You can also select a **face** of a body directly (no sketch) and Extrude it to push or pull the surface. That is a quick edit, though it is less "designed" than editing a sketch.

Press Pull: Extrude's fast cousin

Press `Q` (Press Pull in the Modify panel) and drag a face to push it in or out. Press Pull automatically adapts: on a flat face it extrudes; on a fillet edge it changes the radius; on an offset face it offsets. It is a time-saver for fast edits (but always check the timeline afterward).

TIP

Cut ahead of the extrude. If you want a pocket 5 mm deep, make sure the sketch is on the top face, choose Extrude, and use a **negative** distance or choose the Cut operation, which is automatically selected by Fusion when a profile sits on a body face and you drag inward.

CHAPTER 15

Revolve, Sweep, Loft and friends

The rest of the shape-making tools in the Create panel.

Revolve



Figure 22. A half-profile spun around an axis creates a symmetrical turned part, like what a lathe produces.

1. Draw the profile on one side of a line that will be the axis. The profile must be a closed shape that does not cross the axis.
2. Choose **Revolve**, click the profile, then click the axis (a sketch line, a construction line, an edge, or an origin axis).
3. Set the angle: **Full** (360°), **Angle** (a custom wedge) or **To object**.
4. Pick an operation (Join, Cut, New Body…).

A **Cut** revolve is also how you turn grooves and rings into a cylindrical part.

Sweep and Loft

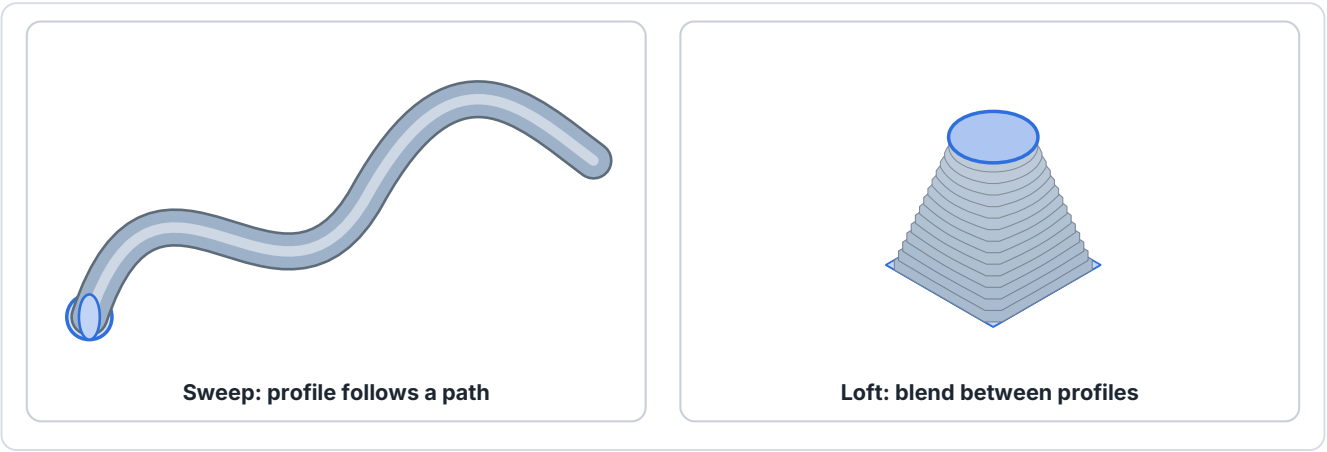


Figure 23. Sweep (left) follows a path; Loft (right) blends two profile shapes.

Sweep

Option	What it does
Profile	The shape to drag. A closed sketch or even a face.
Path	The curve it follows (a sketch curve or chain of edges).
Distance	How far along the path to go (default is the full path).
Taper / Twist	Narrow the profile or twist it along the path.
Orientation	**Perpendicular** keeps the profile facing the path; **Parallel** keeps it fixed in one direction.
Guide Rail	An additional curve that controls the profile's size or shape as it travels.
Type	Single path, path + guide rail, or path + guide surface.

The path and the profile live in two *different* sketches, usually on perpendicular planes. Sketch the path first, then create a plane at the path's start point for the profile (Chapter 20: Plane Along Path).

Loft

Loft blends between two or more profile sketches (or faces). Select them in order. Options:

- **Centerline**: guide the loft along a curve for smoother control of the shape.
- **Rails**: edges that pull the outer shape through the profiles.
- **Edge / Face conditions**: Free, Tangent or Curvature continuity at the ends. Tangent gives smooth joins with existing bodies.
- **Closed loop**: connects the last profile back to the first, such as for a ring.

> **TIP**
> If a loft twists, check that each profile's start point lines up. Dragging the small point markers in the loft dialog fixes most twisting.

Rib and Web

A **Rib** is a thin wall created from a sketch curve (usually a line) extending to the surrounding walls, useful as a triangular brace in a bracket. A **Web** makes a lattice of ribs inside a shell, either based on sketch lines or a pattern. Specify thickness and which side to fill.

Emboss

Emboss lifts or engraves sketch shapes or text on a face, including curved faces. Choose **Emboss** to raise it or **Deboss** to cut it in, set the depth, and optionally add taper. It is perfect for names on handles or logos on lids.

Coil and Pipe

- **Coil**: creates a helix. Choose Revolutions, Height or Pitch, and a section shape (circular, square, triangular, or an outside or inside pitch thread). It is the way to make a spring or a modeled thread.
- **Pipe**: puts a circular or square section along a path with an optional hollow. Great for rails, handles and wire sculptures.

Thicken, Boundary Fill and Derive

- **Thicken** converts a surface into a solid by adding thickness (Chapter 25).
- **Boundary Fill** fills the space enclosed by surfaces and bodies to create a solid, useful for cavities and molds.
- **Derive** imports bodies from another Fusion file and keeps them linked so edits can flow through.

CHAPTER 16

Holes and threads

Almost every mechanical hobby project needs holes. Fusion's Hole and Thread tools save lots of sketching.

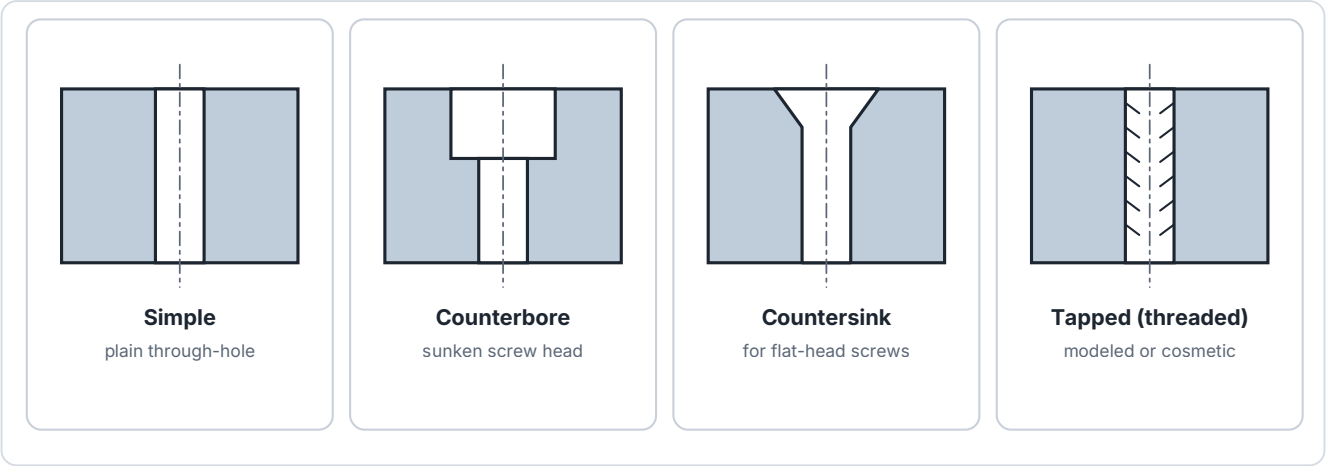


Figure 24. The three hole types in section view, plus a tapped hole.

The Hole dialog

Field	What it does
Placement	**Single Point**: click a face, then position and dimension it. **From Sketch**: pick existing sketch points for many holes at once.
Hole type	**Simple**, **Counterbore** or **Countersink**.
Hole tap type	**Simple** (plain), **Clearance** (loose fit for a screw of a standard size), **Tapped** (threaded) or **Taper Tapped**.
Drill point	Flat or angled (118°) bottom.
Extents	Distance, All (through) or To Object.
Diameter, depth	Typed values, or taken from the standard thread or clearance table.

HOBBYIST ANGLE

For 3D printing, plastic usually shrinks slightly, making a printed hole a little smaller than designed. Add 0.1 to 0.3 mm of clearance to a hole that must take a screw or pin, and test with a small calibration print.

Threads

The **Thread** command works on cylindrical faces (external threads) or holes (internal threads). Key options: **Modeled** (actual 3D geometry, which you need for printed threads) vs cosmetic (just a texture and a label), **Full Length** or a specific length, standard **Thread Type** (such as ISO Metric), **Size**, **Designation**, **Class** and **Direction** (right- or left-hand).

Cosmetic threads are fast and keep file sizes small. For printed bolts and jar lids, use modeled threads and give them extra clearance.

> **WATCH OUT**
> Modeled threads slow Fusion down on big models. Add them near the end of your design, or keep them cosmetic and only model them on export parts.

CHAPTER 17

The Modify menu

Once geometry exists, Modify tools change it. Here is the whole panel.

Tool	What it does	Key
Press Pull	Push or pull faces and edges with automatic behavior.	`Q`
Fillet	Rounds edges.	`F`
Chamfer	Bevels edges.	
Shell	Hollows a solid, leaving walls.	
Draft	Tilts faces for mold release or printing.	
Scale	Resizes a body, uniformly or by axis.	
Combine	Join, Cut or Intersect two bodies (a Boolean operation).	
Offset Face	Moves a face a set distance without a sketch.	
Replace Face	Reshapes a face to match another face or surface.	
Split Face	Divides a face into regions using a curve or plane (for painting or targeting edits).	
Split Body	Cuts one body into two using a plane, surface or sketch.	
Silhouette Split	Splits faces at the outline visible from a chosen direction (mold parting lines).	
Move / Copy	Moves, rotates or copies bodies, faces and components. Offers Free Move, Translate, Rotate and Point to Point.	`M`
Align	Aligns one body or component to another using faces, edges or points.	
Delete	Removes selected bodies, faces or features.	`Del`
Physical Material	Assigns real-world properties (mass, strength).	
Appearance	Assigns a color or texture.	`A`
Manage Materials	Opens the material library.	
Change Parameters	Opens the Parameters table.	
Compute All / Compute Selected	Rebuilds the timeline after a big change.	

Combine: the Boolean operations

When two bodies overlap, **Combine** merges or subtracts them:

- **Join**: merges the target and tool bodies into one.
- **Cut**: subtracts the tool from the target. Check **Keep Tools** if you will reuse the tool shape (such as a tapered cutter).
- **Intersect**: keeps only the shared volume.

> **HOBBYIST ANGLE**
>
> Combine with Cut is how you carve a pocket that exactly fits an object: model the object, make a copy (Move/Copy > Create Copy), enlarge it by your clearance with Scale, then Combine > Cut it out of a holder. Perfect for tool trays and device cradles.

Move/Copy in detail

Mode	Use
Free Move	Drag in any direction with manipulator handles.
Translate along axis	Type X, Y, Z distances and rotation angles.
Point to Point	Click a source point and a target point to snap one onto the other.
Create Copy	Check this box to duplicate rather than move.

CHAPTER 18

Fillet, Chamfer, Shell and Draft

These four finishing tools turn a blocky model into something that looks, feels and prints like a real product.

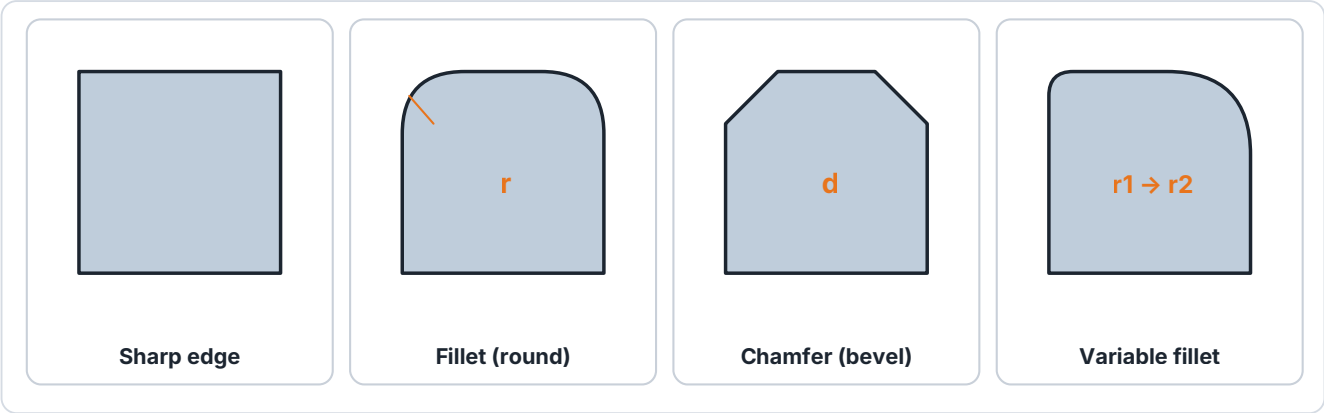


Figure 25. The four edge treatments. Round or bevel the edges you handle most.

Fillet

Press F, select edges (or faces, which rounds all edges of that face) and type a radius. Options:

- **Constant radius** (default), **Chord length** (specify the chord instead of the radius) and **Variable radius** (different radius at each end of the edge).
- **Tangent chain**: automatically picks edges that continue smoothly.
- **Corner type**: Rolling ball (default) or **Setback**, which controls how three fillets blend together at a corner.
- **Continuity**: G1 (tangent) or G2 (curvature) for smoother surfaces.

> **TIP**
>
> **Fillet last.** Place fillets near the end of the timeline. If you fillet early, later sketches and extrusions may stumble over the rounded edges. Also fillet large radii first and small radii after.

Chamfer

A chamfer is a flat bevel. Types: **Equal distance** (same length on both faces), **Two distances** and **Distance and angle**. Chamfers are excellent for the bottom edge of a 3D-printed part, because a 45° chamfer prints cleanly while a vertical edge can squish and flare (an effect called "elephant's foot").

Shell

Select one or more faces to remove (the openings) or select the whole body to make a sealed hollow. Set **Inside Thickness**, **Outside Thickness**, or both, and choose whether the thickness goes inside or outside the current shape. Use a thickness of at least 2× your nozzle width for 3D prints.

Draft

Draft applies an angle to faces, relative to a pull direction. Real molded parts need draft so they release; 3D prints do not, but a slight draft can improve look and help parts pop free of a mold or resin vat. Choose **Plane**, then the faces to draft, then an angle.

Scale

Scale resizes a body from a point. **Uniform** keeps proportions; **Non-uniform** lets you scale X, Y and Z independently. Use it to enlarge an imported model or to create a clearance copy for a cut-out.

Split Body and Split Face

Split Body cuts a body into pieces using a plane or surface. That is how you slice a big model into halves that fit a small print bed. **Split Face** divides a single face so that one region can be offset or textured separately.

CHAPTER 19

Patterns and Mirror

Repetition is where CAD shines. Never copy and paste the same feature twice.

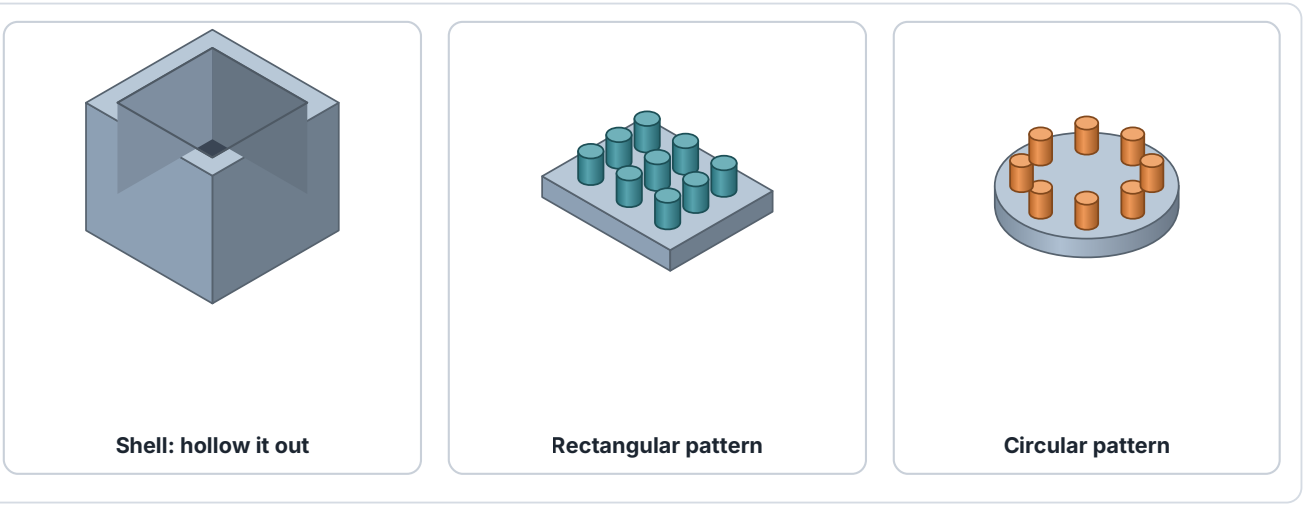


Figure 26. Left to right: Shell, Rectangular Pattern and Circular Pattern.

Pattern types

Pattern	What it does
Rectangular	Repeats along one or two directions. Specify the number and the spacing, with **Extent**, **Spacing** or **Symmetric** distance types.
Circular	Repeats around an axis with a set number and angle. Use **Full** (360°) or **Angle**.
Pattern on Path	Repeats along a curve.

Choose the pattern's **Object Type**: **Features** (repeat timeline features), **Faces**, **Bodies** or **Components**. For a simple row of holes, pattern the hole *feature*; for a row of bolts, pattern the *component*. The key option **Compute Option** (Optimized, Identical or Adjust) controls how Fusion handles patterns that hit different geometry; "Adjust" is slower but most accurate.

Mirror

Mirror copies features, faces, bodies or components across a plane. Draw half a symmetric shape, then mirror it and join. It is simple to do and keeps both halves in sync.

> **TIP**
> Model one half of any symmetric part, put the first sketch on the plane of symmetry, and mirror late in the timeline. You halve the work, and the model stays exactly symmetrical.

Use a pattern for...	Example
Rectangular	Grid of ventilation slots, array of magnet pockets.
Circular	Bolt circle, spokes, scalloped knob edge, fan blades.
On path	Teeth along a curved rail, beads on a necklace.
Mirror	Left and right halves of a handle, symmetric brackets.

CHAPTER 20

Construct: planes, axes and points

Construction geometry gives sketches and features something to attach to when no face is in the right place.

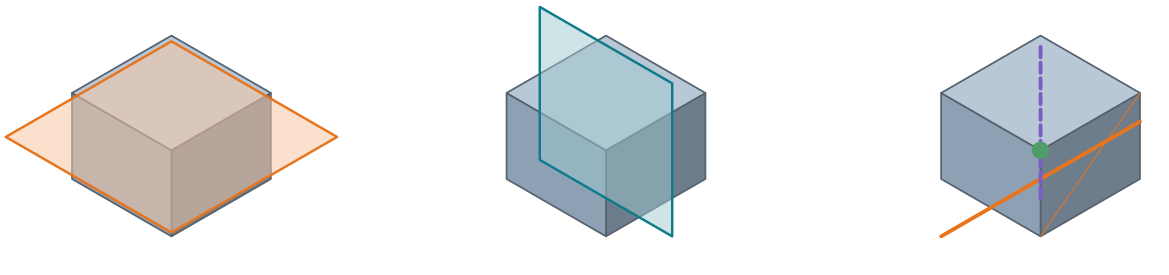


Figure 27. Examples of construction planes, axes and points.

Planes

Command	Creates a plane…
Offset Plane	Parallel to a face or plane at a set distance.
Plane at Angle	Rotated around an edge or axis by an angle (for slanted surfaces such as a phone stand back).
Tangent Plane	Touching a cylinder or sphere.
Midplane	Halfway between two faces or planes.
Plane Through Two Edges	Defined by two edges.
Plane Through Three Points	Defined by three points.
Plane Tangent to Face at Point	Tangent to a curved face at a chosen point.
Plane Along Path	Perpendicular (or at an angle) at a position along a curve. Required for sweeps.

April 2026 update notes added the ability to **pattern** work planes, axes and points (rectangular, circular and path), plus a perpendicular plane from a face or work plane. They also introduced a **User Coordinate System** with a custom origin and axes, stored as an editable timeline feature.

Axes and points

- **Axis Through Cylinder/Cone/Torus**: finds the center axis of a round face for revolves and patterns.
- **Axis Perpendicular at Point** and **Axis Through Two Planes/Points/Edges**.
- **Point at Vertex**, **Point Through Two Edges**, **Point Through Three Planes**, **Point at Center of Circle/Sphere/Torus**, **Point at Edge and Plane**, and **Point Along Path**.

> **TIP**
> Name your construction planes (for example "Lid plane") and hide them once you are done using the light bulb, so the canvas stays clean.

CHAPTER 21

Inspect: measure and analyze

Before you print or cut, check the model. The Inspect panel offers measurement and quality checks.

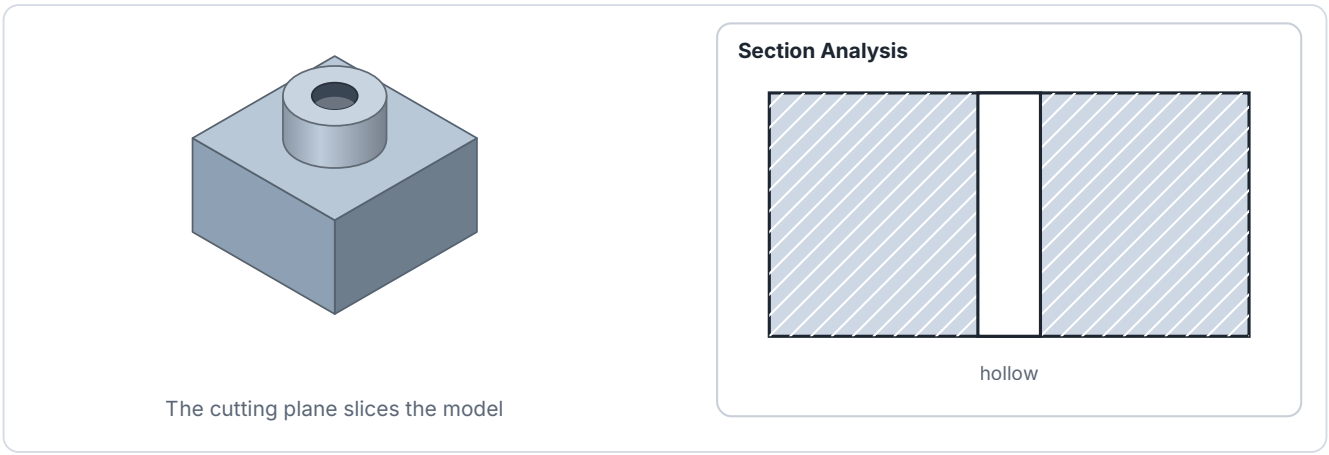


Figure 28. Section Analysis slices the model on a plane so you can check hidden features like walls and cavities.

Tool	What it tells you
Measure `I`	Click faces, edges, points or bodies to read distance, angle, area, length and more. Select two objects to read the distance between them. April 2026 notes say you can now edit the measure selection without clearing it.
Interference	Finds overlapping volumes between bodies. Use it to confirm a lid and box do not collide.
Curvature Comb Analysis	Visualizes the curvature of curves and sketch splines.
Zebra Analysis	Projects stripes on a model to check smoothness of surfaces. Kinks in the stripes mean kinks in the surface.
Draft Analysis	Colors faces by their draft angle.
Curvature Map Analysis	Colors a surface by curvature, highlighting tight spots.
Isocurve Analysis	Shows a grid of lines along a surface to judge its flow.
Accessibility Analysis	Finds areas a tool cannot reach (useful for machining and molding).
Minimum Radius Analysis	Highlights tight corners.
Section Analysis	Cuts the view through the model with an adjustable plane, with an option to measure on the cut.
Center of Mass	Shows the balance point, needed for rotating or balancing parts.

Tool	What it tells you
Display Component Colors	Colors each component differently to spot them in an assembly.

Getting weight and volume

Right-click a body or component and choose **Properties**. If you have assigned a physical material, Fusion shows **mass**, **volume**, **area** and **center of mass**. For 3D printing, the volume divided by 1000 gives cubic centimeters, which you can multiply by the filament's density for a weight estimate. Remember that infill will make the real weight lower.

CHAPTER 22

Insert: bringing in outside content

Fusion can use images, drawings, meshes and standard parts from other places.

Tool	What it does
Insert Derive	Inserts geometry from another design with a link.
Decal	Places a picture (logo, label) on a face for rendering.
Attached Canvas	Places a reference image on a plane. Use it to trace a photo, blueprint or sketch of a part.
Insert Mesh	Brings in an STL, OBJ or 3MF file as a mesh body.
Insert SVG	Imports vector artwork as sketch curves ready to extrude (logos, silhouettes, name plates).
Insert DXF	Imports 2D CAD or laser-cutting files.
Insert McMaster-Carr Component	Searches the McMaster-Carr catalog of real hardware (screws, bearings, extrusions) and inserts 3D models.
Insert Manufacturer Part	Downloads vendor parts.
Insert Fastener	Adds bolts, screws, nuts and washers from the standard content library.
Insert into Current Design	Inserts another Fusion file as a component (the File menu holds Open and Upload as well).

A traced-photo workflow

1. Photograph the object flat-on with a ruler beside it.
2. Choose **Insert > Attached Canvas**, pick a plane, and select the image.
3. Use **Calibrate** to set the scale from the ruler.
4. Trace the outline with splines and lines, then extrude.

April 2026 update notes also say the standard content library gained more than a hundred new component families, which means more fasteners and hardware are ready to insert without searching elsewhere.

CHAPTER 23

Select and selection tools

Selecting the right thing is half of CAD. Fusion offers several ways to refine what you click.

Tool	Behavior
Select	Default selection of any edge, face, body or component.
Selection Priority	Restricts picks to **Edge**, **Face**, **Body** or **Component**. Switch to Body priority when you want to move a whole body rather than a face.
Selection Filters	Lets you toggle specific entity types (vertices, edges, faces, bodies, sketch curves and more) so you only click on those.
Window Selection	Drag a rectangle to select everything inside it. Dragging left to right selects only things completely inside; right to left selects everything the box touches.
Freeform Selection	Draw a lasso around the items.
Paint Selection	Brush over faces to select many quickly.
Selection Sets	Name and reuse a group of selected items.

Selection tips

- Hold `Ctrl` (Windows) / `Cmd` (Mac) to add or remove items; `Shift` for ranges in lists.
- In Fillet and similar dialogs, leave **Tangent Chain** on so one click selects every edge that flows smoothly into it.
- Right-click and choose **Select Other** to pick from a list of everything under the cursor, including hidden or overlapping items.
- Use **Display Settings > Objects Visibility** to hide clutter so you can see what you are picking.

5

PART 5

Beyond solids

Fusion is not limited to blocky parametric parts. This part covers direct modeling, surfaces, T-Spline sculpting, mesh editing, sheet metal and plastic-part tools, each useful for particular kinds of hobby projects.

Chapter 24 · Parametric vs direct modeling 25 · Surface modeling 26 · Form (T-Spline) sculpting 27 · Mesh tools 28 · Sheet metal 29 · Plastic, Utilities, scripts and add-ins

CHAPTER 24

Parametric vs direct modeling

Fusion lets you work either way, even in the same design.

Two philosophies

	Parametric (history-based)	Direct
How it works	Each step recorded in the timeline; edit any step later.	Push, pull and move geometry directly with no recorded history.
Strength	Easy to change dimensions; ideal for families of parts and design intent.	Fast for one-off edits and for imported models with no history.
Weakness	Changing early steps can break later ones.	No timeline to rewind, no parameters to tweak.
Typical use	Enclosures, mechanisms, anything you will revise.	Fixing a downloaded STEP file; quick concept edits.

Capture Design History

In the Browser, right-click the top component or open Document Settings and toggle **Capture Design History**. When it is **on**, features are recorded. When it is **off** (the "direct modeling" setting), edits are applied immediately and the timeline disappears. You can switch back and forth, but turning history off for an existing design removes the timeline, so Fusion warns you first.

> **TIP**
> When you open an imported STEP or other neutral-format file, it arrives as a body with no history. Use **Create > Create Base Feature** to make edits that you can still undo and rollback without recording a full history for the imported geometry.

Useful direct-editing tools

- **Press Pull** to push/pull faces.
- **Move/Copy** on faces to slide a hole or a wall without needing sketches.
- **Offset Face** to grow or shrink a surface.
- **Replace Face** to make a face match another surface.
- **Delete (face)**: removes a face and Fusion heals the gap, which is a fast way to remove fillets or holes from imported models.

CHAPTER 25

Surface modeling

Surfaces are paper-thin skins with no volume. They let you build smooth, complex shapes that are awkward to make with solids, then close them into a solid.

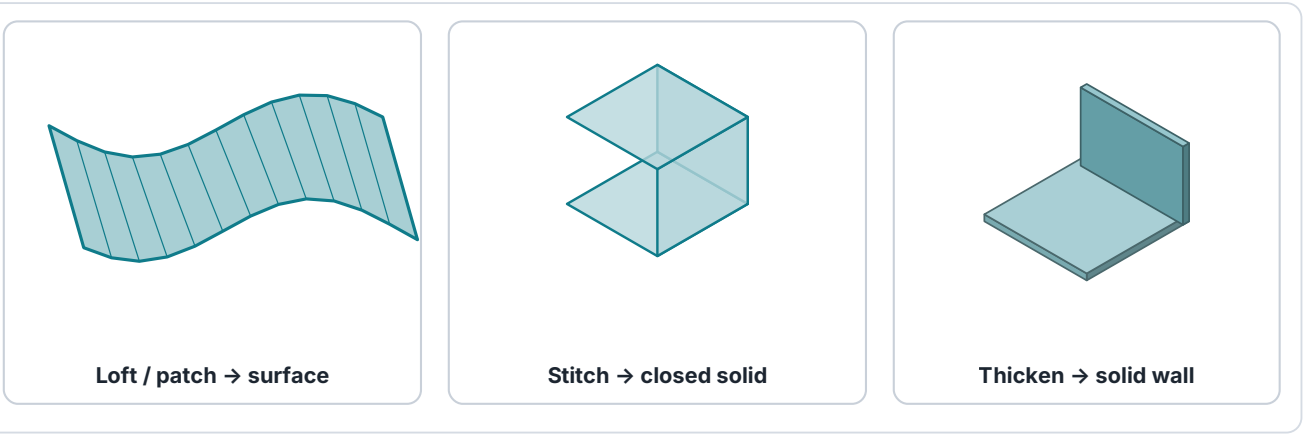


Figure 29. Left: a lofted surface. Middle: surfaces stitched into a closed solid. Right: Thicken turns a surface into a wall.

Why use surfaces?

- Shapes with compound curves (a boat hull, a car body, a headphone shell) that do not come from simple extrudes.
- Repairing or cutting solids with a curved surface (use the surface as a cutting tool in Split Body).
- Complex blends that are easier to control with separate patches.

The Surface tab

Tool	What it does
Extrude, Revolve, Sweep, Loft	Same as the solid versions but create surfaces (no thickness).
Patch	Fills a closed boundary of edges with a surface; handy for capping holes.
Ruled Surface	Creates a surface at an angle and distance from an edge chain.
Offset	Creates a parallel copy of a face or surface at a distance.
Trim	Cuts a surface with another surface or sketch.
Extend	Lengthens a surface beyond its edge.
Stitch	Joins surface patches into one; if they enclose a volume, Fusion makes a solid.
Unstitch	Separates stitched faces.

Tool	What it does
Untrim	Removes a trim to restore the original edges.
Fillet	Rounds edges between surfaces.
Reverse Normal	Flips which side of the surface is "front" (important for stitching and rendering).
Thicken	Gives the surface thickness to make a solid.

A basic surface workflow

1. Create profile sketches or curves.
2. Loft or sweep them into surface patches.
3. Trim and extend patches until their edges meet cleanly.
4. **Stitch** the patches. If the result is watertight, Fusion converts it to a solid.
5. Fillet and shell the solid as usual.

WATCH OUT

If Stitch fails to make a solid, there is a tiny gap between the patches. Increase the **Tolerance** slightly, or inspect the edges: gaps appear as bright edges after Stitch.

CHAPTER 26

Form (T-Spline) sculpting

For smooth, organic shapes like a computer mouse, a figurine base or a stylized helmet, Fusion offers a sculpting-style workspace called Form.

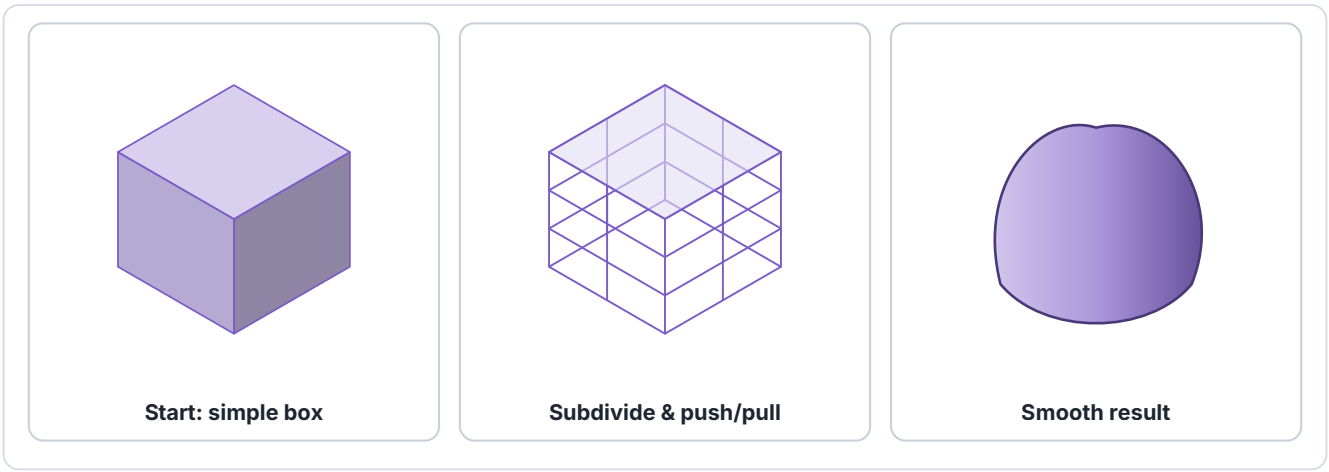


Figure 30. Start with a simple box, subdivide it into a control cage, and push and pull points; the surface smooths itself automatically.

What is a T-Spline?

A T-Spline is a smooth surface controlled by a coarse "cage" of faces, edges and vertices. Moving the cage's points reshapes the smooth surface beneath. It is like working with modeling clay but with the precision of CAD. Click **Create > Create Form** to enter the Form environment, where a green contextual tab with its own tools appears. Click **Finish Form** when done and the result becomes a regular solid or surface body.

Form tools

Tool	What it does
Box, Plane, Cylinder, Sphere, Torus, Quadball	Starting primitives. Quadball is a sphere made of quads; ideal for rounded organic starting shapes.
Face, Pipe	Create a custom face by clicking points, or a pipe shape along a path.
Extrude, Revolve, Sweep, Loft	Create form geometry from sketches.
Edit Form	The key tool. Select a vertex, edge or face and drag the on-screen handles (move, rotate, scale).
Insert Edge	Adds an edge loop to give you more control points where you need detail.

Tool	What it does
Subdivide	Splits a face into more faces.
Insert Point	Adds a single point on an edge.
Merge Edge, Bridge, Fill Hole, Weld Vertices	Join and close parts of the cage.
Crease / Uncrease	Makes selected edges sharp or smooth.
Bevel Edge, Slide Edge	Adds bevels or slides edge loops.
Match, Interpolate, Flatten	Align edges to surfaces and curves, smooth along a curve, or flatten an area.
Thicken	Gives a form sheet thickness (for hollow shells).
Mirror (symmetry)	Edit one side while the other updates. A toggle in the dialog.
Display modes	Switch between smooth, box (cage) and a blend of both so you can see the underlying control points.

A tip for beginners

Start with as *few* faces as possible. Beginners tend to subdivide too much and end up with lumpy surfaces. Add detail only where you need it. Turn on symmetry and always keep the cage a clean grid of four-sided faces.

> **HOBBYIST ANGLE**
>
> Form is wonderful for custom ergonomic handles, game controller shells, bike parts and sculpted holders. Combine a Form body with the solid tools later: finish the Form, then add screw holes and mounting bosses with ordinary parametric features.

CHAPTER 27

Mesh tools

Mesh files (STL, OBJ, 3MF) are made of thousands of tiny triangles. They are what 3D printers and 3D scanners use, and Fusion can edit them.

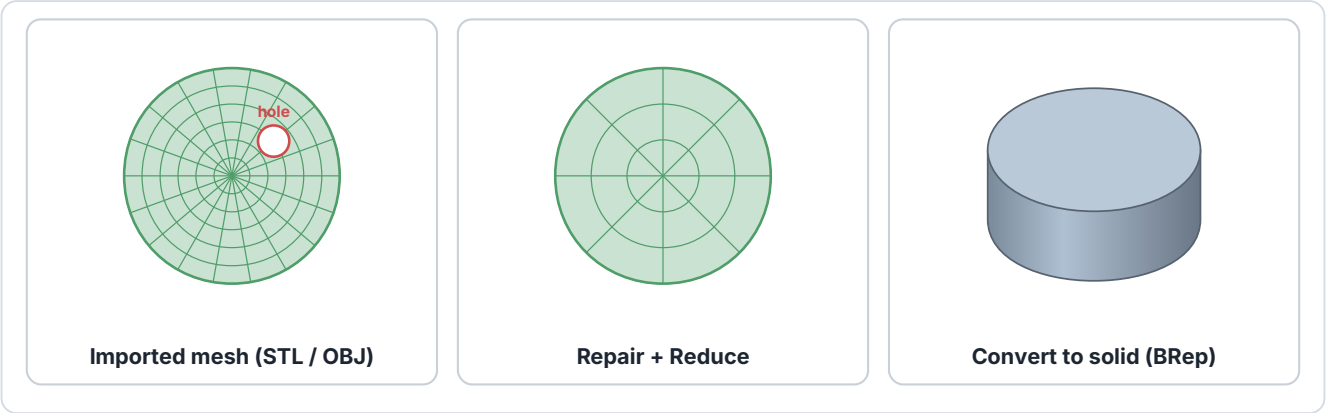


Figure 31. A mesh with a hole is repaired and simplified, then converted into a true solid (BRep) body that can be edited with all of Fusion's modeling tools.

Mesh vs solid

- A **mesh** is a shell of triangles. It has no "features," just surface shape. Great for printing and scanning, hard to edit precisely.
- A **BRep (boundary representation) solid** has true curves and faces. Great for editing and measuring.

The Mesh tab

Tool	What it does
Insert Mesh	Brings an STL, OBJ or 3MF file into your design.
Reduce	Cuts the triangle count (faster, smaller) while keeping the overall shape.
Remesh	Rebuilds the triangulation for evenness.
Repair	Fixes holes, flipped normals and other defects. April 2026 notes added a **One Touch Fix** option that runs a combination of repair operations in one step.
Erase and Fill	Removes selected triangles and fills the gap smoothly.
Plane Cut	Cuts a mesh with a plane and optionally caps it.
Combine	Merges meshes.
Reverse Normals	Flips triangle orientation.

Tool	What it does
Shell / Smooth	Hollows or smooths a mesh.
Mesh Section Sketch	Creates sketches from slices through a mesh, to rebuild it as CAD.
Generate Face Groups	Groups triangles into surface-like regions so the mesh can be converted into a solid with clean faces.
Convert Mesh	Converts a mesh into a BRep solid or Form (usually via face groups).

Common mesh jobs

1. **Fix a downloaded model:** Insert Mesh > Repair > check that it is watertight > export for printing.
2. **Add a custom mount to a scanned object:** Insert Mesh, then sketch around it (projected edges are available through sections) and model the mount as a regular solid around it.
3. **Split a big model for a small printer:** Plane Cut with capping.
4. **Edit an STL as CAD:** Reduce > Generate Face Groups > Convert Mesh. Results depend greatly on the shape.

WATCH OUT

Converting complicated organic meshes to solids can produce huge or failed bodies. Reduce the mesh first, and try the Faceted conversion if smooth fails. For text and logos, SVG import is usually a better route.

CHAPTER 28

Sheet metal

If you build enclosures, brackets or panels from folded sheet (or laser-cut and bent acrylic), the Sheet Metal tab designs the folded part and unfolds it to a flat pattern.

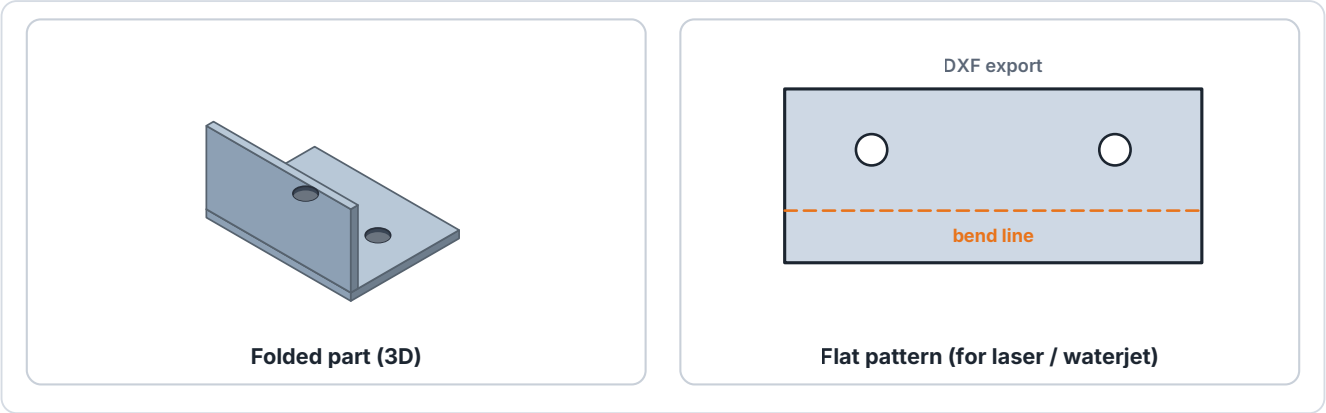


Figure 32. A folded bracket and its flat pattern, which can be exported as DXF for laser, plasma or waterjet cutting.

Sheet metal rules

A **Sheet Metal Rule** sets the material thickness, bend radius, bend relief and corner treatments. Choose or create a rule in the Browser's Sheet Metal Rules or the Sheet Metal tab. Different metals (aluminum, steel) and thicknesses use different rules, and the flat pattern accounts for how metal stretches around a bend.

Key tools

Tool	What it does
Flange	Adds a bent wall to an edge of a sheet.
Contour Flange	Extrudes a sketched profile into a sheet with bends.
Bend	Bends a flat region along a line.
Hem, Rip, Corner Closure	Fold over an edge, split a seam, and clean up corners. April 2026 notes added automatic **Corner Closure**, following the active rule.
Unfold / Refold	Temporarily flatten bends to add cut-outs that cross them, then fold back.
Create Flat Pattern	Makes a flat version with bend lines and dimensions.
Export Flat Pattern as DXF	Saves a cutting file (note: export formats may be limited on personal licenses).

Hobbyists with access to online laser-cutting or waterjet services can design a part, export the flat pattern, and have bent metal parts delivered. You can also use the Sheet Metal tools to design cardboard and plywood boxes that fold flat.

CHAPTER 29

Plastic, Utilities, scripts and add-ins

The last two Design tabs, plus the extension mechanisms, round out the toolbox.

Plastic tab

The Plastic tab has features that speed up designing enclosures made by injection molding or 3D printing. Typical tools include **Boss** (screw posts), **Rest** (small locating features), **Grill** (vent slots), **Lip and Groove** (where lid and base meet), **Snap Fit**, and **Rule** settings that define default wall thickness and draft. Even if you never mold a part, the same features make good 3D-printed boxes.

Feature	Hobby use
Boss	Screw post inside a case for self-tapping screws or heat-set inserts.
Lip and Groove	Interlocking edge so the lid sits aligned.
Snap Fit	Printed latches that hold a lid without screws.
Grill	Ventilation for electronics.

Utilities tab

- **3D Print**: sends the selected bodies to your slicer or saves a print file (Chapter 40).
- **Scripts and Add-Ins**: opens the dialog where you run and edit scripts, enable add-ins, and create your own.
- Other helpers such as design-checking and workflow utilities. The exact set can change between updates.

Scripts, add-ins and the API

Fusion has a programming interface (API) usable from Python or C++. Hobbyists use it to automate repetitive tasks or generate geometry from math, such as a parametric gear, a spiral vase or a perforated panel. The Autodesk App Store and community sites offer free add-ins for things like gear generators, text engraving and printable hinge makers.

> **TIP**
> April 2026 update notes mention an **Autodesk Assistant** capability that can write and run Fusion API scripts from plain-language requests. Availability may depend on your plan, but it lowers the bar to automation for people who do not code.

6

PART 6

Assemblies and motion

Real projects have more than one part. This part explains how to put components together, how joints define motion, how to test mechanisms, and how to use standard hardware and design variants.

Chapter 30 · Assemblies and joints
31 · Motion studies and contact
32 · Fasteners and standard parts
33 · Variants and configurations

CHAPTER 30

Assemblies and joints

An assembly is a group of components positioned relative to each other. Joints are the rules that connect them.

Grounded and free components

A new component is free to move. **Ground** one component (usually the base) to anchor the assembly: right-click it in the Browser and choose Ground. Everything else is then positioned relative to the grounded part with joints.

Joint vs As-built Joint

Command	Use
Joint J	Pick two points (such as the center of a hole and the center of a pin) and the joint pulls the components together. Components *move* to satisfy it.
As-built Joint	Creates a joint where components already are, without moving them. Use when the parts are already modeled in position.
Joint Origin	A pre-placed anchor point on a component that you can reuse when joining.
Rigid Group	Locks several components together as one unit.

The seven joint types

Joint	Allows	Example
Rigid	No motion at all	A lid screwed down permanently
Revolute	Rotation around one axis	Door hinge, wheel on an axle
Slider	Translation along one axis	Drawer, sliding lid
Cylindrical	Rotation and sliding on the same axis	Screw jack, piston with spin
Pin-Slot	Rotation about one axis plus sliding along another	Pin in a slot, rolling wheel on rail
Planar	Sliding on a plane plus rotation about its normal	Puck on a table
Ball	Rotation in all directions around a point	Ball joint, joystick

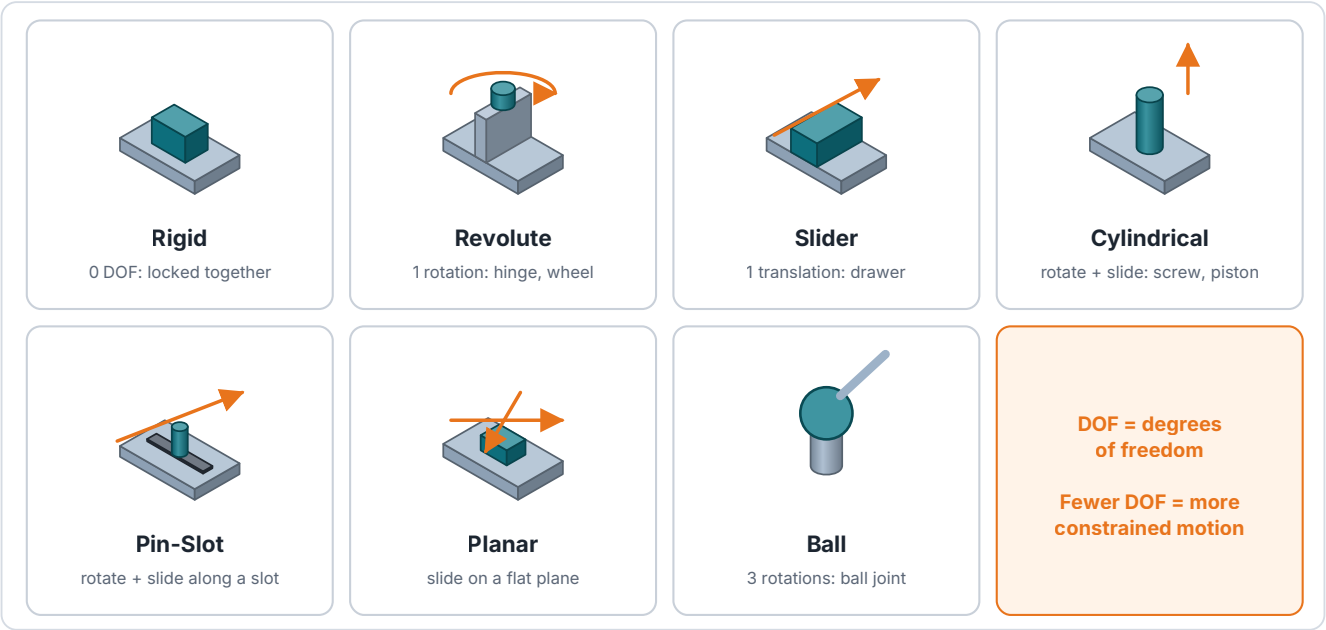


Figure 33. Fusion's joint types. DOF means degrees of freedom: the ways the joined component can still move.

Making a joint, step by step

1. Press `J` or choose **Assemble > Joint**.
2. Click a snap point on the first component. Hover over faces and edges: Fusion shows a small origin marker at hole centers, circle centers and edge midpoints.
3. Click the matching point on the second component. They move together.
4. Choose the joint **Type** and adjust **Offset** or **Angle** if needed. Use **Flip** if a part is facing the wrong way.
5. Check the **Motion** tab to set limits (such as a hinge that only opens 110°).
6. Click OK, then drag the component to test the joint.

WATCH OUT

If a joint seems to do nothing, check that the components are *components*, not just bodies. Joints only work between components, one of the main reasons for the organization discipline in Chapter 7.

Joint limits

On a Revolute or Slider joint you can set minimum and maximum values (Rest, Minimum, Maximum) so the part cannot move beyond them. That is how you stop a hinge opening 360° or a drawer sliding off its rails.

CHAPTER 31

Motion studies and contact

Once joints are in, you can test the mechanism.

Drive Joints and Motion Link

- **Drive Joints** (in the Assemble menu) lets you set a joint to a value by dragging a slider or typing, showing the full range of motion.
- **Motion Link** ties two joints together by a ratio, for gears (one revolute turning another at 2:1), rack and pinion, and cam-like mechanisms.

Contact sets

By default, components can pass through each other when you move them. **Enable Contact Sets** (and **Enable All Contact**) makes selected parts collide, so a drawer stops at the cabinet instead of passing through it. Use it for simple mechanisms; it is not a full physics simulation.

Motion Study

A **Motion Study** (set up in the Simulation workspace) records joint positions over time to create simple animations and to check for interference during movement. Within the Animation workspace you can then create a video.

Check	How
Does it move as intended?	Drive Joints; drag the component.
Does it hit anything?	Enable Contact Sets or run Interference at several positions.
Do parts line up?	Use Section Analysis on the closed position.

HOBBYIST ANGLE

Perfect for hinges, flip-top lids, robot arms, camera sliders and print-in-place mechanisms. Rehearse the motion in Fusion before burning filament.

CHAPTER 32

Fasteners and standard parts

Real projects rely on screws, nuts, bearings and magnets. Fusion can bring in accurate models of these.

Insert Fastener

On the **Insert** panel, **Insert Fastener** opens a library of standard hardware (ISO, ASME and others). Pick a hole and Fusion suggests screw sizes that fit it; choose length, head type and whether to add a nut and washer. The fastener arrives as a component with joints, ready to move.

McMaster-Carr and manufacturer parts

Use **Insert McMaster-Carr Component** to search real catalog items (such as a bearing or aluminum extrusion) and insert accurate 3D models. Use **Insert Manufacturer Part** for vendor-specific models. Real-part models make sure your design fits hardware you can actually buy.

Check the stack

April 2026 notes mention a **Fastener Stack Analysis** in the Stack Analyzer that checks threaded fasteners for insufficient engagement and mismatched thread designations, types or sizes, which is useful when a screw looks right but is too short to grip.

Hardware	Hobby tip
Machine screws (M3)	Use a clearance hole of ~3.4 mm in plastic and a tapped or heat-set insert hole in the mating part.
Heat-set inserts	Model the hole slightly smaller than the insert's outer diameter per the supplier's datasheet.
Neodymium magnets	Model a pocket 0.1 to 0.2 mm larger than the magnet, with a thin cover layer to hide it.
Bearings (608, 625…)	Insert from McMaster-Carr or build a simple component; keep to the datasheet's bore and outer diameter.

CHAPTER 33

Variants and configurations

Making several versions of a design (small, medium, large) is a common hobby need.

Configurations (paid feature)

The **Configure** panel, in plans that include it, lets you define multiple variations (sizes, colors, with/without a feature) inside one file, managed through a **configuration table**. Autodesk lists configurations as not included in personal use, and April 2026 notes describe a Design Extension feature called **Configuration Rules** that restricts or drives configuration values. Not available for personal users, but good to know.

A free alternative: parameters + copies

1. Define user parameters for everything that varies (`width`, `height`, `wall`).
2. Drive all dimensions from those parameters.
3. For each variant, change the parameters, then **File > Export** the STL (or save a version named "Large").
4. Or use **Save As** to make a copy of the file for each variant.

This simple method is how most hobbyists produce size variants of a holder or bracket, and it keeps everything in one tidy parametric model.

> **TIP**
> Keep a "master" design that you never export from, and always **Save As** before changing parameters, so you can return to the standard dimensions.

7

PART 7

Presenting your design

A finished model deserves a good picture. This part covers the Render, Animation and Drawing workspaces, which turn a design into photo-real images, exploded-view videos and dimensioned plans.

Chapter 34 · Render 35 · Animation 36 · Drawings

CHAPTER 34

The Render workspace

Rendering makes a realistic picture of your model with materials, lighting and shadows. It is great for sharing projects and checking how colors and finishes will look.



Figure 34. Appearance swatches. The same shape looks completely different with different materials, and Fusion's library has hundreds.

Switching to Render

Open the workspace selector and choose **Render**. Autodesk describes it as producing photo-realistic images with controls for lighting, materials and decals, using local or cloud computing; cloud rendering is not included in personal-use licenses, but local rendering is.

The render tools

Tool	What it does
Appearance `A`	Opens the material library. Drag a material onto a body, face or component. Right-click > Edit to change color, roughness and reflectivity.
Scene Settings	Controls the environment (a lighting picture or HDRI), brightness, background (solid color, transparent or an image), the ground plane, camera, and exposure. April 2026 notes say scene settings, including camera projection, can now be saved as defaults.
Decal	Applies a picture (logo, label, QR code) to a face with position, scale and rotation controls.
Texture Map Controls	Adjusts how wood grain, carbon fiber or other textures are scaled and rotated.
In-canvas Render	Shows a live ray-traced preview right in the 3D view as you work. Great for quickly checking material choices.
Capture Image	Saves a high-resolution render to a file. Choose size, quality and background.
Render Gallery	Browse and compare saved renders.

Getting a good render in five steps

1. Give each body an **appearance**. Even simple plastic, metal and wood make a huge difference.
2. Open **Scene Settings** and choose an environment that suits the object. A neutral studio setting is safe.
3. Turn on **In-canvas Render** and orbit to find a flattering angle.
4. Adjust **exposure**, the **depth of field** and, if useful, the **focal length**.
5. Click **Capture Image**, choose the resolution and let it finish. Longer renders produce cleaner images.

> **TIP**
> Render with a **transparent background** if you plan to place your product on a website or a poster, and save as PNG.

Appearances for 3D-printed parts

If you want your render to match the real print, search the library for "plastic" and pick a matte finish in your filament color. Adding a faint visible layer texture with a bump map makes the picture convincingly "printed".

CHAPTER 35

The Animation workspace

Show how something goes together. Fusion makes exploded views and animated videos from your assembly.

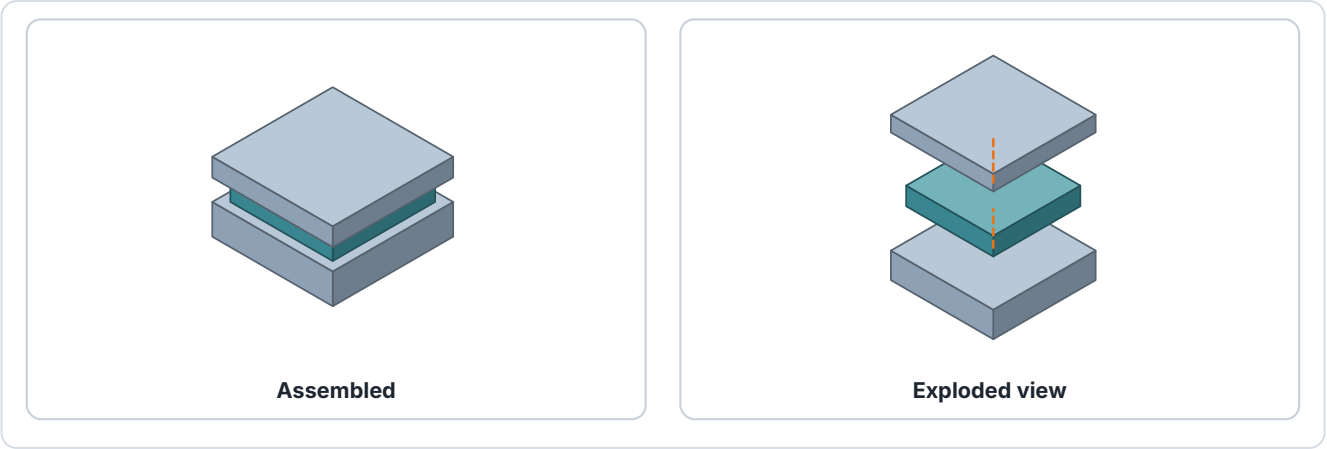


Figure 35. The same assembly in its normal state and as an exploded view, ready to be animated.

Key concepts

- **Storyboard**: the timeline at the bottom, made of **views**. Each view is a snapshot of component positions, visibility, camera and so on.
- **Auto Explode**: automatically spreads out components, either one level (the top-level components) or all levels. Drag the slider to set the distance.
- **Manual Explode**: select components and drag a handle along an axis for exact control.
- **Transform Components**: move or rotate parts between storyboard views to create assembly motions.
- **Show/Hide**, **Callouts** and **Annotations**: add labels and visibility changes.
- **Publish Video**: render the storyboard to a movie file.

Steps to a quick assembly video

1. Switch to the Animation workspace.
2. Click **Auto Explode: All Levels**, then adjust the slider.
3. Rearrange the stack of parts as needed with Manual Explode.
4. Press play in the storyboard to see the animation of the parts flying apart and back together.
5. Click **Publish**, choose resolution and file format, and save.

Exploded views are also useful as pictures to include in instructions for a kit you give away, or for a sheet of assembly steps for a project you plan to share.

CHAPTER 36

Drawings

A technical drawing is a flat sheet that shows dimensions and views, for a machine shop, a laser-cutting service or your own workshop notes.

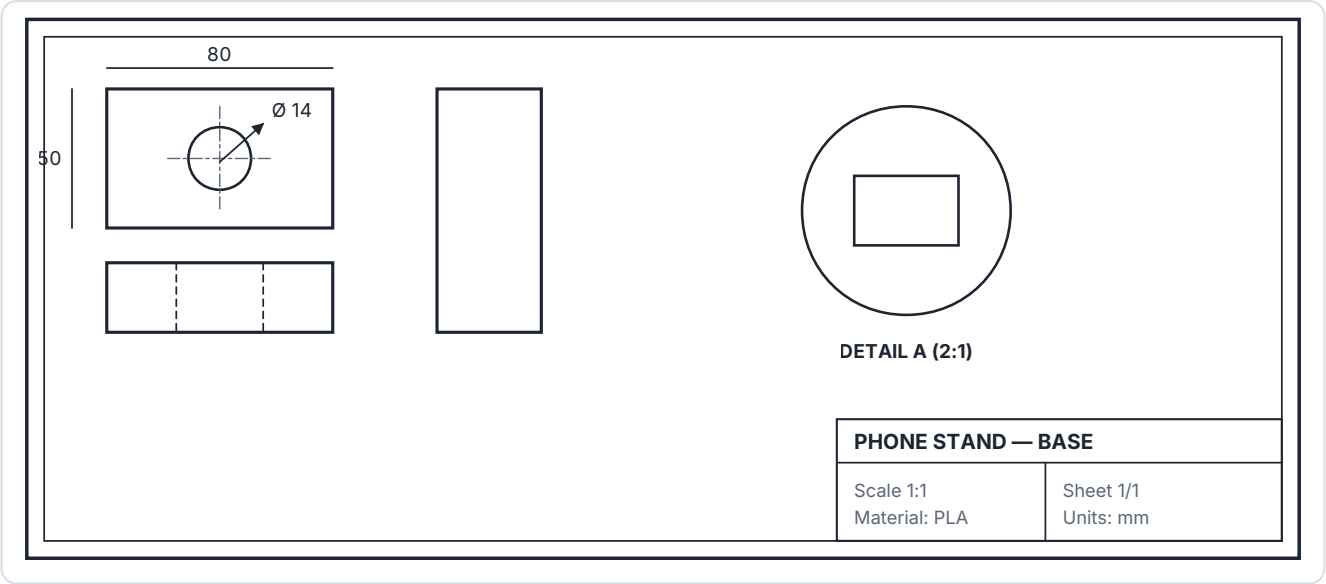


Figure 36. A typical Fusion drawing: front, top and side views with dimensions, a detail view and a title block.

Making a drawing

1. In the Design workspace, choose **File > New Drawing > From Design** (or switch to the Drawing workspace).
2. Pick a standard (ISO, ASME), units and sheet size, and click OK.
3. Click to place the **Base View**; set orientation and scale. Add **Projected Views** by dragging from the base view.
4. Add dimensions, centerlines, hole callouts, and notes.
5. Export to PDF (or other format) for printing or sending.

Drawing tools

Tool	What it does
Base View / Projected View	Primary and derived orthographic views.
Section View	A cut-through view showing the inside.
Detail View	A zoomed circle of a small area.

Tool	What it does
Break View	Shortens a long part in the view.
Auxiliary View	A view perpendicular to an angled face.
Dimension, Ordinate Dimension	Linear, angular, radial and diameter dimensions; ordinate dimensions measured from a datum.
Hole & Thread Notes	Automatic callouts that read the hole feature's size.
Centerlines, Center Marks	Symbolic markings for round features.
Text, Leader, Symbols	Annotations, surface finish and weld symbols.
Tables	Tables, such as a parts list or configuration table. A bill of materials is not included in personal use.

Drawings are **associative**: if you change the model, the views and dimensions update. Autodesk lists 2D drawings and documentation as limited in the personal-use version, so check what you can export. At minimum you can usually produce a PDF for your own use.

HOBBYIST ANGLE

Even if you only 3D print, a drawing makes a handy reference when you build jigs, or if you have a local machine shop cut a part for you. It forces a final check of key dimensions.

8

PART 8

Analysis and making

Fusion can test, optimize and manufacture what you design. This part covers simulation, generative design, CAM for CNC machines, 3D printing preparation and electronics, and notes what the free personal-use license limits.

Chapter 37 · Simulation 38 · Generative design 39 · Manufacture (CAM) 40 · 3D printing 41 · Electronics

CHAPTER 37

Simulation

Simulation asks: "Will it break, bend or overheat?" It uses finite element analysis (FEA), where the model is divided into tiny pieces and the physics is solved on each.

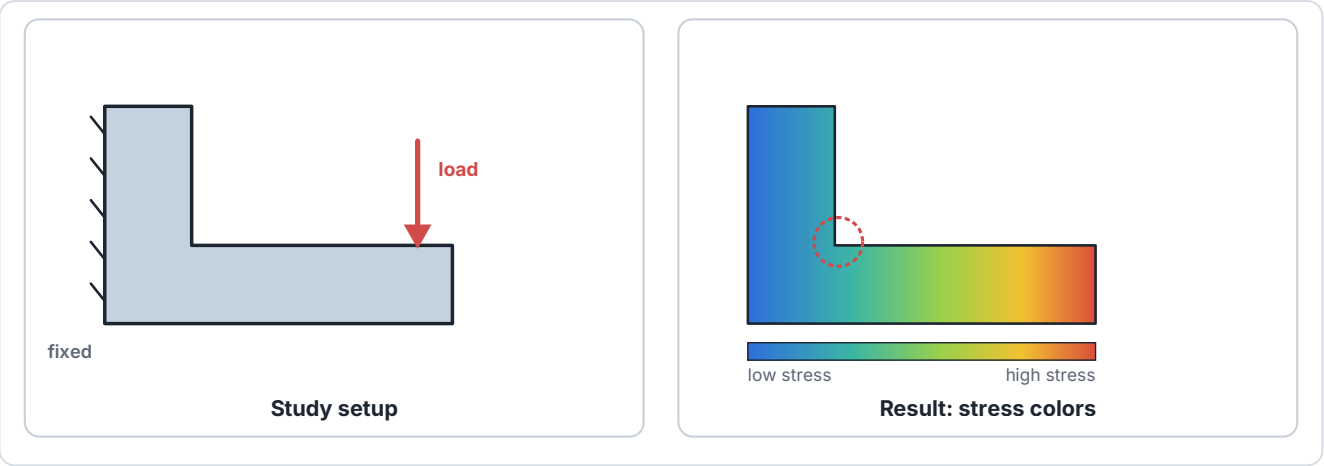


Figure 37. A bracket fixed at one end with a load at the other (left) and the stress result colored from low (blue) to high (red).

Autodesk describes the Simulation workspace as setting up studies to test your design using FEA, so you can see how it performs under loads and conditions and compare alternatives. In the personal-use license, simulation is limited and cloud simulation is not included, so what you can run depends on your account. Check what is available in your own installation.

Study types in Fusion

Study	Question it answers
Static Stress	How much does it deform and where is the stress under a steady load?
Modal Frequencies	At what frequencies does it naturally vibrate?
Thermal	How does heat flow and what are the temperatures?
Thermal Stress	What stresses does temperature change cause?
Structural Buckling	At what load will a slender part buckle?
Nonlinear and Event simulation	Large deformations, impacts and time-based loads (advanced tiers).
Shape Optimization	Where can material be removed while still meeting strength targets?
Injection Molding	How does plastic fill a mold? (extension)

Which study types are available depends on your plan and may change; hobbyists most commonly use Static Stress and sometimes Modal and Thermal.

A static stress study in six steps

1. Switch to the **Simulation** workspace and choose **New Study > Static Stress**.
2. Assign a **Material** to each body (PLA, ABS, aluminum, and so on).
3. Add **Constraints**: where the part is held (fixed faces, pins).
4. Add **Loads**: forces, pressures, gravity or torque.
5. Check the **Mesh** (the tiny elements), adding refinement near stress concentrations.
6. Click **Solve**. View results such as **Von Mises stress**, **Displacement** and **Safety Factor**.

Reading the results

- **Safety Factor** greater than 1 means the material should not yield; for something that matters (a hook, a mount) aim for 2 to 4 or more.
- **Red hotspots** usually appear at sharp inside corners. Adding a fillet lowers the stress there.
- **Displacement** shows how much it bends. A part can be "safe" yet too flexible.

> **WATCH OUT**
>
> 3D-printed parts are **anisotropic**: they are much weaker between layers than along them. A simulation that treats printed plastic as uniform can be over-optimistic. Use generous safety factors and test physically.

Simplify

The **Simplify** environment lets you make changes for simulation only, such as removing small holes, fillets and logos, without changing your real model. It makes the mesh simpler and the solve faster.

CHAPTER 38

Generative design

Instead of drawing the part, you describe what it must do, and Fusion proposes shapes, often organic and lightweight, that meet your goals.

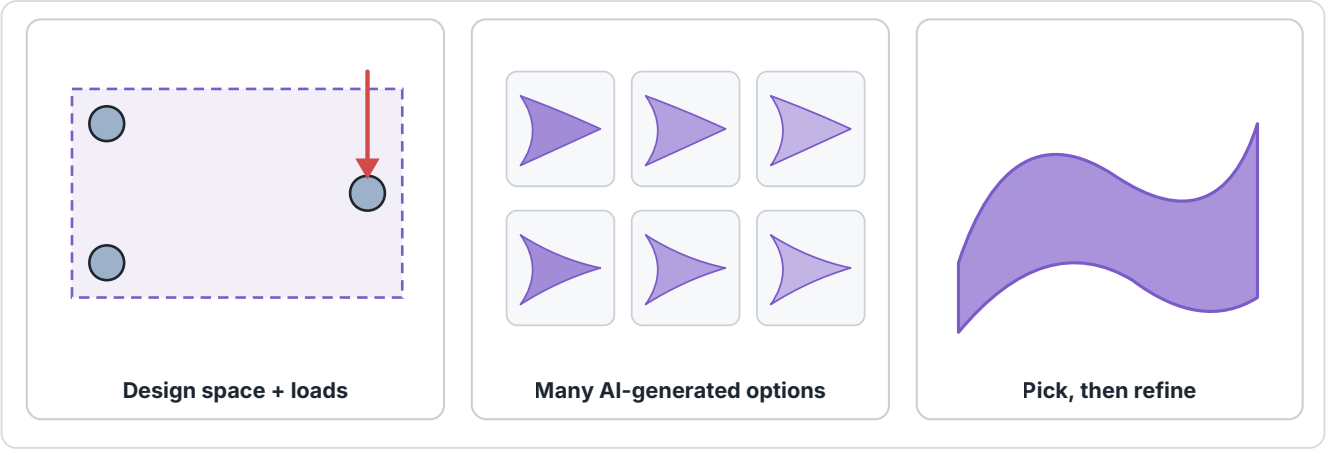


Figure 38. You define a design space, the parts to keep, and loads; the software generates many candidate shapes for you to compare.

How it works

1. **Preserve geometry**: the bits that must stay (bolt holes, mounting surfaces).
2. **Obstacle geometry**: regions that must stay empty.
3. **Starting shape** (optional): the allowed volume.
4. **Structural constraints and loads**: where the part is fixed and the forces it must carry.
5. **Objectives**: for example, minimize mass or maximize stiffness, with a target safety factor.
6. **Manufacturing method**: additive (3D printing), 2-axis or 3-axis or 5-axis milling, die casting, or unrestricted.
7. **Materials**: pick one or several.
8. **Generate**: compare results in a table and chart, then export one as a body to refine.

Generative design uses cloud computing and, in many plans, cloud credits. Autodesk's personal-use comparison page does not address it, so check what your account allows before relying on it.

> **HOBBYIST ANGLE**
>
> For hobby use it is a fun way to design drone frames, bike brackets, camera mounts and other light, strong 3D-printed parts. The results look unusual and can print without supports if you choose the additive constraint.

CHAPTER 39

Manufacture (CAM)

CAM turns your model into instructions for a machine. For hobbyists, that mostly means CNC routers and mills, though Fusion also covers laser, plasma, waterjet, turning and 3D printing.

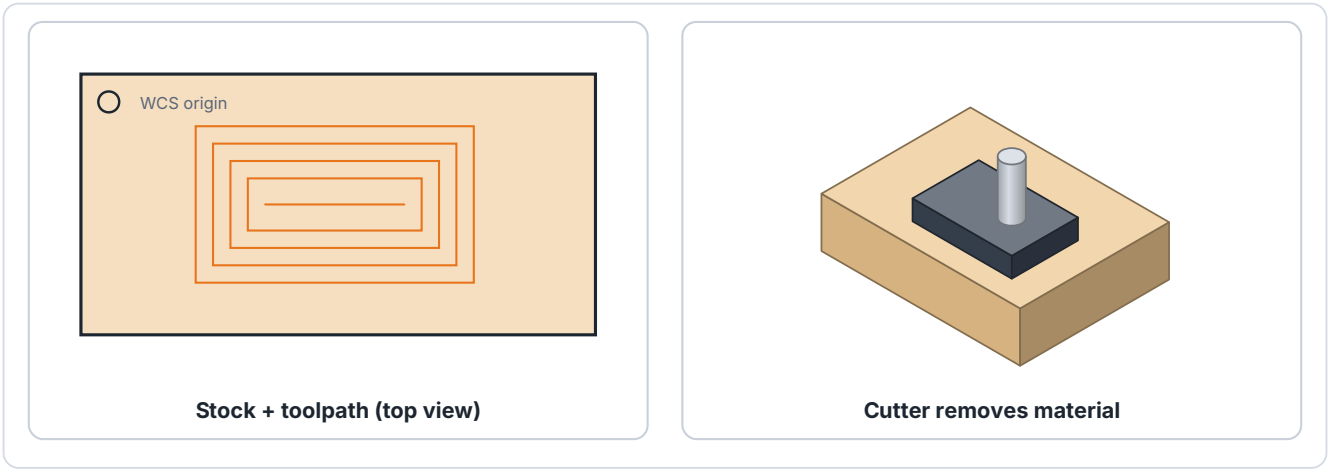


Figure 39. A setup defines the stock and coordinate system; toolpaths tell the cutter where to travel.

Autodesk describes the Manufacture workspace as creating toolpaths for subtractive processes such as machining and turning, and for additive processes such as 3D printing. In the personal-use version, CAM is limited and basic compared to the full toolkit: one third-party summary describes it as basic 2-axis and 3-axis machining without advanced multi-axis. Autodesk's page says toolpath generation has limited functionality, so confirm exactly what your license allows.

The workflow

1. **Setup**: choose the machine type (milling, turning, cutting), the **stock** (raw material), and the **work coordinate system** (the zero point).
2. **Tools**: pick cutters from the tool library or add your own (diameter, flutes, feeds and speeds).
3. **Toolpaths**: create operations: first roughing, then finishing.
4. **Simulate**: watch the cutter remove material and check for collisions and gouges.
5. **Post process**: convert to G-code in your controller's dialect (GRBL, Mach3, Carbide Motion and many others).
6. **Run** on the machine, starting slowly the first time.

Common toolpaths

Toolpath	Used for
Face	Flattening the top of the stock.
2D Adaptive Clearing	Fast roughing of pockets with constant tool load.
2D Pocket	Clearing a flat-bottomed pocket.
2D Contour	Cutting around an outline (profile).
Drill / Bore / Circular	Holes of different kinds.
Engrave / Trace	Following a line for text and artwork.
Thread Milling	Cutting threads. April 2026 notes add Automatic mode that detects Fusion-defined threads.
3D Adaptive / Pocket Clearing	Roughing sculpted shapes.
3D Parallel, Scallop, Contour, Pencil	Finishing curved surfaces and cleaning corners.

Other Manufacture tabs

- **Milling** and **Turning** for subtractive machining.
- **Additive**: print setup, orientation, supports and arrangement (April 2026 notes mention an arrangement criteria table and body presets).
- **Fabrication**: laser, plasma and waterjet cutting from flat patterns.
- **Inspection**: probing.

WATCH OUT

Cutters are dangerous and mistakes are expensive. Run simulation, check the zero point, use eye and ear protection, and keep your hand on the stop button during the first run.

CHAPTER 40

Preparing for 3D printing

Most hobbyists slice in a separate program. Fusion prepares the model for that slicer and can send it over directly.

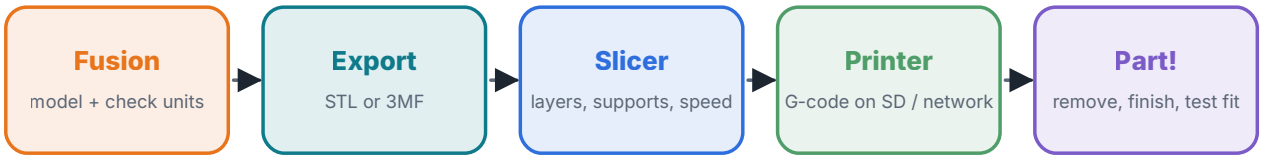


Figure 40. The path from model to printed part.

Sending a model to the slicer

1. Select the body or component to print. For a multi-part design, activate and export one part at a time.
2. Choose **Utilities > 3D Print** (or **File > 3D Print**). In the dialog, pick the target (Send to a Print Utility such as your slicer, or Save as a file).
3. Choose the file format: **STL** (universal) or **3MF** (modern; carries units, color and multiple objects).
4. Set the refinement: higher gives smoother curves but larger files. High is fine for most prints.
5. Open in your slicer (PrusaSlicer, Cura, Bambu Studio, OrcaSlicer and others), set the layer height, infill and supports, and print.

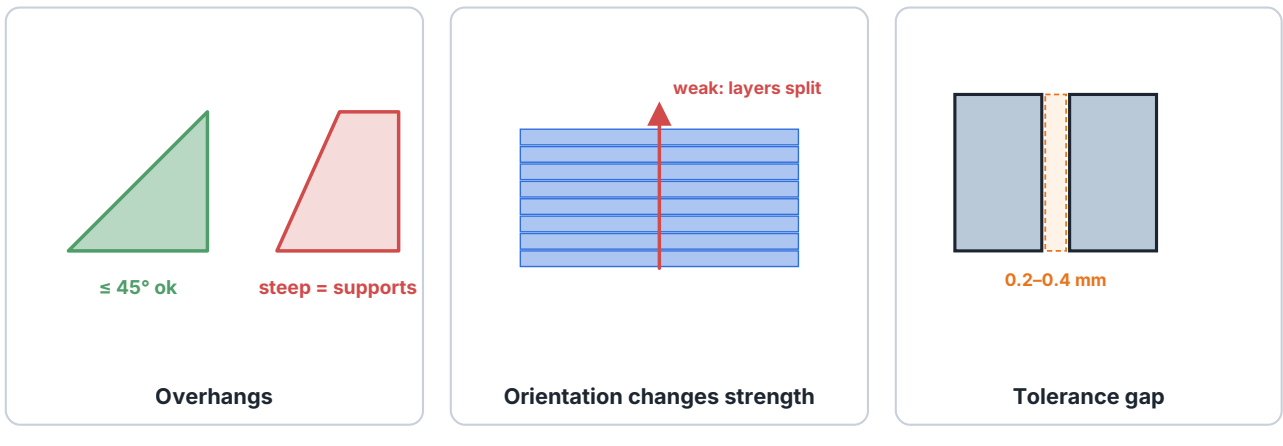


Figure 41. Three ideas that matter for printed parts: overhang angle, orientation and clearance gaps.

Printability checklist

Check	Guideline
Wall thickness	At least two nozzle widths (about 0.8 to 1.2 mm for a 0.4 mm nozzle).
Overhangs	Up to about 45° from vertical usually prints without supports.
Bridges	Short flat spans (under about 5 to 10 mm) usually print fine.
Clearance between moving parts	Around 0.2 to 0.4 mm for FDM printers; test on your own machine.
Holes	Add 0.1 to 0.3 mm to the diameter for screws and pins.
Elephant's foot	Chamfer the bottom edge about 0.4 mm.
Flat on the bed	Choose a large, flat face for the bottom. Rotate the model in the slicer.
Layer direction	Orient parts so forces run along layers, not between them.

The Manufacture workspace's **Additive** tab can also arrange parts on a build plate, set up supports, and generate print settings for selected printers, although the free license limits some of these features.

CHAPTER 41

Electronics

Fusion includes a circuit design environment, so you can design a PCB that fits the case you are modeling.

Autodesk describes the Electronics workspace as creating and editing schematics, 2D and 3D PCB designs, electronic components and component libraries, with libraries shared with EAGLE. The personal-use license includes limited PCB and electronics design (Autodesk's page says *limited*; check the current limits in your account).

Step	What happens
Schematic	Place parts and wire them together on a drawing sheet.
Board outline	Define the PCB shape, possibly from a Fusion sketch.
Place and route	Position components on the board and draw copper traces.
3D PCB	View the board as a 3D model and place it in a case assembly to check fit.
Manufacturing files	Export Gerbers for a PCB service.

The ECAD/MCAD link means moving a connector in the case moves it on the board, and vice versa. April 2026 notes mention a new **Electronics Browser** for managing schematic sheets, modules and instances.

> **HOBBYIST ANGLE**
> Even simple projects benefit: design the box around a real PCB model so that switches, USB ports and LEDs line up the first time.

9

PART 9

Everyday hobby use

Now put it all together. This part explains the features that make Fusion easy to learn, walks through four complete beginner projects, and ends with cheat sheets, troubleshooting, a four-week learning plan and a glossary.

Chapter 42 · Why Fusion is easy to learn 43 · Four starter projects 44 · Designing for printing, cutting and CNC 45 · Shortcuts cheat sheet 46 · Troubleshooting 47 · Learning roadmap 48 · Glossary

CHAPTER 42

Why Fusion is easy to learn

Fusion is powerful, but several design choices make it friendlier than its feature list suggests.

Ten features that lower the learning curve

Feature	Why it helps beginners
Timeline with rollback	You can see every step you took, fix an early mistake without redoing everything, and replay the build as a time-lapse.
Live previews	Dialogs show the result as you type or drag, and nothing is applied until you click OK.
Color-coded sketches	Blue means "not locked down yet", black means "fully defined". The feedback is visual and instant.
Automatic constraints	Fusion guesses horizontal, vertical, tangent and coincident relationships while you draw.
Typed values with Tab	Precise dimensions come from the keyboard as you draw, without a separate dimensioning step.
Marking menu and command search	Frequently used commands are under your cursor, and any command can be found by typing its name.
Contextual tabs	When you are in a sketch or in the Form environment, the toolbar shows only the relevant tools, hiding hundreds of irrelevant ones.
Cloud versions	Every save is recoverable. Experimentation is safe.
Everything in one program	You do not need to learn separate programs for modeling, rendering and preparing a print.
Learning resources built in	The Home page links to tutorials and courses, the Help menu is searchable, and the forum and community are active.

The Autodesk Assistant

Autodesk's April 2026 update notes describe an **Autodesk Assistant** that can perform modeling steps from plain-language requests, including extrude, fillet, chamfer, hole, shell, split, patterns, materials and sketch dimensions, and answer geometry questions like volume and surface area. It can also create manufacturing setups and toolpaths, write and run Fusion API scripts, and (for commercial users, with a monthly allowance) generate render-style images from the current view. Features like these can change

between releases and plans, so try the assistant in your own install before depending on a particular capability. Treat any assistant-generated geometry the way you would a tutorial: check the timeline and the numbers.

Official ways to learn

- **Autodesk Learn**: free on-demand courses at autodesk.com/learn, including a "Learn Fusion for CAD in 90 minutes" course listed for 2026, plus courses on drawings and generative design.
- **Fusion Help**: help.autodesk.com has a "New to Fusion" section with a walkthrough that builds a first design, and detailed pages for each workspace.
- **The Autodesk Fusion blog**: step-by-step articles, such as beginner sketching guides and monthly "What's New" posts.
- **Community forums**: ask questions, share files, and search for answered problems. For personal-use users, the forum is also the support channel.

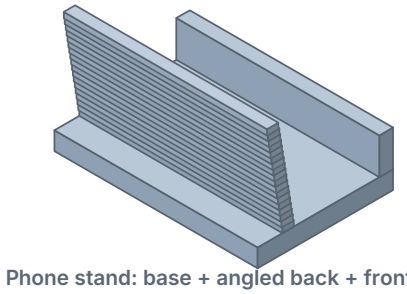
> **TIP**
> Learn by remaking things on your desk. A pen cup, a phone stand, a cable clip: simple objects you can measure with a ruler teach the same skills as complex projects, and the real thing sits there to check your work.

CHAPTER 43

Four starter projects

Follow these in order. Each introduces new tools while reusing earlier ones. Dimensions are suggestions; change them to suit your own phone, desk or printer.

Project 1: Phone stand



Phone stand: base + angled back + front lip

Tools used

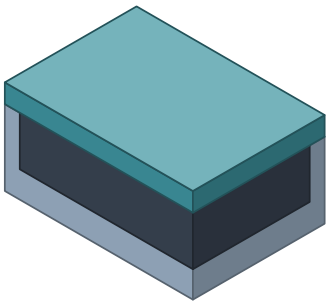
Sketch: rectangle, dimension
Extrude: base plate
Sketch on a face + Extrude: back
Angle via Construct > Plane at Angle
Fillet: soften every edge
Hole / cut-out: charging cable slot

Figure 42. A phone stand: a base, an angled back and a front lip.

1. **New design.** Check units are mm. Rename the component "Phone Stand".
2. **Base.** Sketch a center rectangle 90 × 70 on the floor plane. Extrude 6 mm.
3. **Back support.** On the side plane, draw a slanted four-sided shape 8 mm thick that starts at the back edge of the base and leans back, rising 70 mm. Dimension its angle to the base as 75°. Extrude it **Symmetric** 90 mm with **Join**.
4. **Front lip.** Sketch a 90 × 6 rectangle at the front on the top face and Extrude 14 mm so the phone does not slide forward.
5. **Charging slot.** Sketch a 14 mm wide slot in the middle of the base, extrude with **Cut** through all.
6. **Finish.** Press `F` and fillet all outer edges 2 mm. Fillet the lip 1.5 mm.
7. **Export.** Utilities > 3D Print > save 3MF. Slice with 15% infill.

Skills practiced: sketching on faces, Extrude with Join and Cut, Fillet, symmetric extrusion, parameters (try making `phone_thickness` a parameter).

Project 2: Box with a pin hinge



Box with removable lid and a pin hinge

Tools used

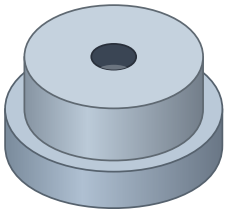
Components: Body, Lid, Pin
Shell: hollow out the box
Offset plane + sketch for the lid
Revolute joint: hinge motion
Pattern: vent slots or magnets
Appearance: material per part

Figure 43. A box and lid with a pin hinge, modeled as separate components.

1. **Create components** named Box, Lid and Pin. Activate Box.
2. **Box body.** Sketch 100 × 70, extrude 40 mm. Use **Shell** on the top face with a 2.4 mm thickness to hollow it.
3. **Lid.** Activate Lid. Sketch on the top rim; project the outer edge; Offset inward by 0.3 mm for clearance to make a lip. Extrude 3 mm for the plate and a 4 mm tall inner lip.
4. **Hinge barrels.** At the back edge, create circles of 8 mm diameter along the back, alternating box and lid, each 12 mm long (use a rectangular pattern).
5. **Pin.** Activate Pin. Sketch a 3.2 mm circle on the end of a barrel and extrude through all barrels. Leave a 0.2 mm gap on the lid barrels.
6. **Joints.** Press `J`: Revolute joint between Lid and Pin; Rigid between Pin and Box. Set limits 0° to 110°.
7. **Test.** Drive Joints to check the motion. Use **Inspect > Interference** to confirm no clash.

Skills practiced: components, Shell, Project and Offset, patterns, joints with limits, interference checking.

Project 3: Custom knob



Custom knob with grip pattern and D-shaft hole

Tools used

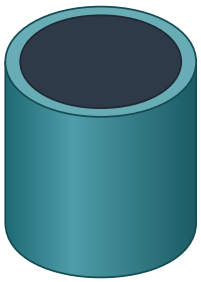
Sketch profile + centerline axis
Revolve: full 360° body
Hole sketch with a flat (D-shaft)
Circular Pattern: scallops around the rim
Fillet: comfortable edges
Combine: cut grip grooves

Figure 44. A knob made with Revolve, with a pattern of grip scallops and a D-shaped hole.

1. **Profile.** On the front plane, draw a half-profile: bottom width 20 mm radius, top 16 mm radius, 25 mm tall, with a slightly curved side. Add a vertical centerline on the left as the axis.
2. **Revolve** the profile 360°.
3. **Shaft hole.** Sketch on the bottom: a 6 mm circle with one side trimmed flat to make a D shape (flat at 5.4 mm across). **Extrude** with Cut 12 mm deep.
4. **Grip scallops.** Sketch a 5 mm circle on the top-face rim, Extrude with Cut down the side. Use **Pattern > Circular** with 12 copies around the center axis.
5. **Edges.** Fillet top edge 1.5 mm.
6. **Mark.** Use **Sketch > Text** plus **Emboss** (or a shallow Extrude Cut) to engrave a small arrow on top.

Skills practiced: Revolve, construction axes, circular patterns, cutting, text.

Project 4: Pen cup with a name



Pen cup: revolve or extrude, then shell

Tools used

Circle sketch + Extrude (or Revolve)
Shell: open the top, 2–3 mm wall
Fillet bottom & rim
Emboss / text for a name
Parameters: height & diameter
Export STL → slicer

Figure 45. A simple pen cup: a cylinder with a thick floor and an engraved name.

1. Create user parameters: `height` = 95 mm, `dia` = 80 mm, `wall` = 2.4 mm.
2. Sketch a circle with diameter `dia`, extrude to `height`.
3. **Shell** the top face, with thickness `wall`. Check the floor is thick enough (change the bottom thickness separately if needed).
4. Add a 1.5 mm fillet to the rim and 2 mm to the base.
5. Sketch text on a plane tangent to the cylinder. Use **Emboss** on the curved face, 0.6 mm deep.
6. Change `height` and `dia` to make a mini version. Watch everything update.

Skills practiced: user parameters, Shell, Fillet, Emboss on curved faces, and the power of a parametric model.

KEY IDEA

After these four projects, you have used Sketch, Extrude, Fillet, Shell, Revolve, Pattern, Components, Joints, Parameters and Export. That is the core of everyday Fusion use.

CHAPTER 44

Designing for printing, cutting and CNC

A model that looks perfect on screen can still fail on a machine. These rules of thumb bridge the gap.

3D printing (FDM) rules of thumb

Topic	Advice
Orientation	Model parts in their printing orientation if you can. Put the flat face on the XY plane.
Tolerance parameter	Create `clearance` as a parameter (start at 0.3 mm) and use it everywhere parts must fit.
Fillets	Add fillets to inside corners to reduce stress concentration and increase strength.
Shrinkage	Some materials (ABS, nylon) shrink more than PLA; test fit and adjust.
Split big parts	Use Split Body, then add alignment pins or a dovetail so the pieces mate.
Text	Keep embossed text at least 0.8 mm tall strokes, with at least 0.4 to 0.6 mm raise.
Print-in-place	Moving parts need 0.3 to 0.5 mm gaps between surfaces.

Laser cutting and sheet goods

- Design 2D profiles as sketches; export as DXF (or SVG) via the sketch's right-click menu or from a flat pattern.
- Measure your real material thickness with calipers. Plywood labeled 3 mm is often 2.8 or 3.2 mm. Make it a parameter.
- Account for **kerf** (the width of material the laser removes, around 0.1 to 0.3 mm) with an offset.
- For finger joints, pattern slots with a width equal to the thickness minus a small fit tolerance.

CNC routing and milling

- Inside corners cannot be sharper than your cutter radius. Fillet inside vertical corners with at least the cutter radius (or add "dog-bone" relief for square-fit joints).
- Think about workholding: tabs, clamps, and double-sided tape need room.
- Keep deep, narrow pockets to a depth of about three times the cutter diameter.
- Use the CAM Simulate step to check for collisions before running.

General habits: model what you can measure with calipers, test-print just the tricky fit region before the whole part, work in millimeters, and name things as you go.

CHAPTER 45

Shortcuts cheat sheet

Default keys in Fusion. Shortcuts can be customized in preferences, and macOS users replace `Ctrl` with `Cmd`.

Navigation and general

Action	Shortcut
Orbit	`Shift` + middle mouse drag
Pan	Middle mouse drag
Zoom	Scroll wheel
Fit to screen	Double-click middle mouse
Undo / Redo	`Ctrl`+`Z` / `Ctrl`+`Y`
Save	`Ctrl`+`S`
Cancel command	`Esc`
Marking menu	Right-click
Quick-access / search palette	`S`
Delete selected	`Delete`

Sketch

Tool	Shortcut
Line	`L`
2-Point Rectangle	`R`
Center Diameter Circle	`C`
Sketch Dimension	`D`
Trim	`T`
Offset	`O`
Project	`P`
Toggle construction line	`X`

Tool	Shortcut
Move / Copy	M

Modeling

Tool	Shortcut
Extrude	E
Press Pull	Q
Fillet	F
Hole	H
Joint	J
Measure	I
Appearance	A

TIP

Fusion lets you change shortcuts. Look in your profile Preferences, or right-click a command in the toolbar and look for a keyboard-shortcut option (exact menu names vary by version).

CHAPTER 46

Troubleshooting

The problems almost every beginner meets, and how to get out of them.

Symptom	Likely cause and fix
Extrude does not highlight a profile	The sketch is not closed or has overlapping lines. Use Show Profile, Trim extras, and look for small gaps.
Fillet fails	The radius is too big for the geometry. Try a smaller radius, fillet the larger ones first, or change the selection order.
Shell fails	Thickness is too big for thin or curved areas. Reduce it, or fillet before shelling.
Red exclamation marks in the timeline	An early edit broke a later feature. Roll back to the first red item; re-select the missing face or edge in its dialog.
Cannot select what I want	Check selection priority; hide bodies in the way; use Select Other from the right-click menu.
Sketch appears in the wrong place	It was created on a different plane or in a different component. Check the Browser and component activation.
Joint does nothing	Make sure the parts are components, and that one is not grounded in the way.
The model is slow	Hide unused bodies, turn off effects in Display Settings, simplify or suppress threads, and group timeline items.
STL imports at the wrong size	Units mismatch. Use Scale, or change the import unit.
Print is too tight or loose	Adjust your `clearance` parameter and re-export.
Everything sits in one body	Use Split Body and Move to Component (or redo with Extrude's New Component operation).
Cannot export a format	Personal-use licenses limit some formats. Try STL, 3MF, or STEP as available.
Cloud sync is slow	Check your connection and the job status icon at the top; wait for uploads to finish before closing.

KEY IDEA

When you are lost: press **Esc**, double-click the middle mouse button to fit the view, and look at the timeline. Nine out of ten problems become obvious once you see which step created them.

CHAPTER 47

Learning roadmap

A four-week plan of about five hours a week.

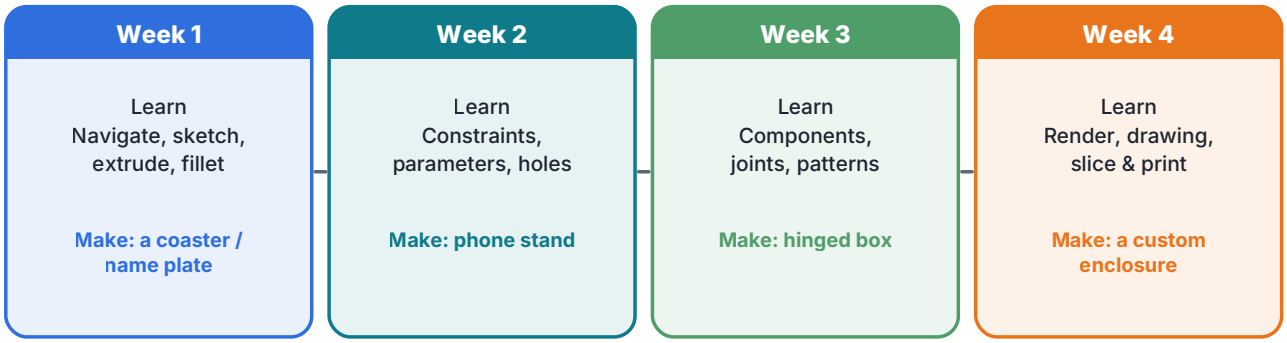


Figure 46. A gentle four-week plan from first sketch to first print.

Week 1: Move and make

- Install, sign in, set units and orientation.
- Practice navigation: orbit, pan, zoom, ViewCube.
- Draw twenty simple sketches. Practice Line, Rectangle, Circle, Trim and Dimension.
- Extrude and Fillet. Make a coaster with a name on it.

Week 2: Precision

- Learn constraints and aim for black sketches.
- Build the phone stand, then change one parameter and watch it update.
- Use Hole, Chamfer, Shell and Pattern.

Week 3: Parts that work together

- Create components and name them.
- Make the hinged box. Add joints, limits, and test motion.
- Try Revolve and Sweep with the knob and a handle.

Week 4: Share and make

- Render an image and make an exploded animation.
- Export a print file, slice it, and print a fit test.

- Design something you actually need: a drawer organizer, a bracket, a replacement knob.

Beyond

Choose a direction: **3D printing** (tolerances, print-in-place), **woodworking** (joinery, CNC), **electronics** (PCB plus enclosure), **organic design** (Form), or **engineering** (simulation, generative design). Each builds on the same foundation.

CHAPTER 48

Glossary

Appearance
The visual look of a body: color and finish. Separate from physical material.

Assembly
A design with multiple components positioned relative to each other.

BRep
Boundary representation: a solid defined by true faces and edges (as opposed to a triangle mesh).

Body
A single solid or surface in a design.

CAD / CAM / CAE
Computer-aided design, manufacturing and engineering (analysis).

Canvas
The 3D view area. Also, an inserted reference image.

Chamfer
A flat bevel on an edge.

Component
A part container that can have its own bodies, sketches and joints.

Constraint
A geometric rule in a sketch, such as parallel or tangent.

Construction geometry
Reference lines, planes, axes and points that guide design but do not make solids.

DOF
Degrees of freedom: the ways a component can still move.

Extrude
Push a 2D profile into 3D along a direction.

FDM
Fused deposition modeling: filament 3D printing.

FEA
Finite element analysis: the numerical method behind simulation.

Feature
One modeling operation stored in the timeline.

Fillet
A rounded edge.

Flat pattern
A sheet-metal part unfolded flat.

G-code
Machine instructions produced by CAM or a slicer.

Ground
Fix a component in space so others move relative to it.

Hub
Your top-level cloud space in Fusion.

Joint
A relationship that defines how two components can move.

Kerf
The width of material removed by a cutting process.

Loft
A smooth blend between two or more profiles.

Mesh
A surface made of triangles (STL, OBJ, 3MF).

Origin
The point (0, 0, 0) where the three base planes meet.

Parametric
Driven by editable values and a recorded history.

Parameter
A named value used in sketches and features.

Profile
A closed region in a sketch.

Slicer
Software that cuts a model into layers for 3D printing.

STEP / STL / 3MF
Common exchange formats: STEP for solid CAD data, STL for triangle meshes, 3MF for modern print files.

Sweep
Move a profile along a path to make a shape.

T-Spline
A smooth, editable surface controlled by a cage of points.

Timeline
The ordered list of features at the bottom of the screen.

Toolpath
The route a cutter travels, generated in CAM.

ViewCube
The interactive cube for standard views.

WCS
Work coordinate system: the zero point used by a CNC program.

APPENDIX

Sources and notes

Product facts about licensing, workspaces and the April 2026 update come from Autodesk's own pages where possible. The rest is general Fusion knowledge presented as a learning aid; menus and defaults vary between versions.

1. Autodesk, "Fusion for Personal Use: The Best Free CAD Tool You Should Consider Upgrading" (August 2026), autodesk.com/products/fusion-360/blog/?p=90315
2. Autodesk, "Free, limited functionality for hobbyist users" (Fusion for personal use page), autodesk.com/products/fusion-360/personal
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5. Autodesk Fusion blog, "Sketches in Fusion" and "How To Create and Modify Sketch Geometry in Fusion 360", autodesk.com/products/fusion-360/blog/
6. University of Edinburgh uCreate Studio, Fusion 360 interface walkthrough, ucreatestudio.is.ed.ac.uk/node/2718
7. Fabbaloo, "Fusion 360 Licensing Change: Reactions And Autodesk Retreat", fabbaloo.com
8. Autodesk Learn, "Learn Fusion for CAD in 90 minutes (2026)", autodesk.com/learn/ondemand/course/learn-fusion-for-cad-in-90-minutes-2026
9. Community reports on personal-use limits (for example the 10-document cap) are from user forums and should be verified in your own account.

LAST WORD

Fusion rewards curiosity. Open it, break something, press undo, and try again. Every expert started with a wobbly rectangle. Happy making.